

Triangular minimality for planar Coulomb renormalized energy

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Abstract

We prove the Sandier–Serfaty conjecture: the triangular lattice of covolume one minimizes planar Coulomb renormalized energy over all admissible curl-free fields with a unit uniform background. Combined with Bétermin and Sandier’s asymptotic formula, this also proves the Brauchart–Hardin–Saff conjecture for the linear term of optimal ordered-pair logarithmic energy on the unit two-sphere. The proof uses a direct Voronoi-cell comparison with rigorous interval arithmetic.

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1 Introduction

The Coulomb interaction in two dimensions is logarithmic. An infinite configuration of equal point charges against a uniform neutralizing background has an infinite electrostatic energy, even in a bounded region containing a charge. Subtracting the logarithmic self-energy at each charge and then taking energy per unit area gives the renormalized energy studied here. Its proposed minimizer is the triangular lattice.

1.1 The energy and the main results

For $R > 1$, let $K_R = [-R, R]^2$. Fix smooth cutoffs $\chi_R \in C_c^\infty(\mathbb{R}^2)$ satisfying

$$0 \leq \chi_R \leq 1, \quad \text{supp } \chi_R \subset K_R, \quad \chi_R = 1 \text{ on } K_{R-1}, \quad \sup_{R>1} \|\nabla \chi_R\|_\infty < \infty. \quad (1.1)$$

Let $\Lambda \subset \mathbb{R}^2$ be locally finite and simple, so every point has multiplicity one, and put $\nu_\Lambda = \sum_{p \in \Lambda} \delta_p$. The admissible class $\mathcal{A}_1$ consists of locally integrable vector fields E such that

$$\text{div } E = 2\pi(\nu_\Lambda - 1), \quad \text{curl } E = 0, \quad \sup_{R>1} \frac{\nu_\Lambda(B_R)}{|B_R|} < \infty, \quad (1.2)$$

where the equations hold in distributions and B_R is the disk of radius R centered at the origin. These conditions imply $E \in L_{\text{loc}}^q(\mathbb{R}^2; \mathbb{R}^2)$ for every $1 \leq q < 2$: locally subtract the finitely many Coulomb singularities and a smooth field with the prescribed constant divergence; the remaining field has harmonic components. Section 2 supplies the details. Consequently the same class results if local L^q integrability is required for any one, or for all, of these exponents.

For $E \in \mathcal{A}_1$ define

$$W(E, \chi_R) = \lim_{\eta \downarrow 0} \left[\frac{1}{2} \int_{\mathbb{R}^2 \setminus \bigcup_{p \in \Lambda} B(p, \eta)} \chi_R |E|^2 dx + \pi \log \eta \sum_{p \in \Lambda} \chi_R(p) \right], \quad (1.3)$$

$$W(E) = \limsup_{R \rightarrow \infty} \frac{W(E, \chi_R)}{|K_R|}. \quad (1.4)$$

The local puncture limit exists and is finite. The outer limsup is initially allowed to take either infinite value. The puncture limit is taken first, and the cutoff layer has the fixed width in (1.1) throughout.

Identify $\mathbb{R}^2$ with $\mathbb{C}$ when convenient. The triangular lattice of covolume one is

$$\Lambda_\Delta = \sqrt{\frac{2}{\sqrt{3}}} \left(\mathbb{Z}(1, 0) + \mathbb{Z} \left(\frac{1}{2}, \frac{\sqrt{3}}{2} \right) \right). \quad (1.5)$$

Let h_Δ be the mean-zero periodic distributional solution of $-\Delta h_\Delta = 2\pi(\delta_0 - 1)$ on $\mathbb{R}^2/\Lambda_\Delta$, and set $E_\Delta = -\nabla h_\Delta$.

Theorem 1.1. *For every cutoff family satisfying (1.1) and every field $E \in \mathcal{A}_1$,*

$$W(E) \geq W(E_\Delta).$$

The triangular value is finite, and $\inf_{E \in \mathcal{A}_1} W(E) = W(E_\Delta)$.

In particular $W(E) = -\infty$ is impossible in this class. The inequality includes fields of energy $+\infty$. No separation, periodicity, or Voronoi-cell bound is imposed on an arbitrary competitor.

The finite-volume statement used to prove Theorem 1.1 has its own exact normalization. For an integer $n \geq 2$, write $T_n = \mathbb{R}^2/(\sqrt{n}\mathbb{Z}^2)$. Given distinct points $a_1, \dots, a_n \in T_n$, let h have mean zero and satisfy $-\Delta h = 2\pi(\sum_{i=1}^n \delta_{a_i} - 1)$ on T_n . Define

$$\mathcal{W}_n(h) = \lim_{\eta \downarrow 0} \left[\frac{1}{2} \int_{T_n \setminus \bigcup_{i=1}^n B(a_i, \eta)} |\nabla h|^2 dx + \pi n \log \eta \right]. \quad (1.6)$$

Theorem 1.2. *For every integer $n \geq 2$ and every configuration of n distinct points on T_n , the potential in (1.6) satisfies*

$$\mathcal{W}_n(h) \geq nW(E_\Delta).$$

This is a lower bound on square tori. It does not assert equality on any particular square torus. Constant harmonic fields increase the energy by $n|c|^2/2$; the required orthogonality follows by integrating the periodic gradient, as proved in (2.12).

The planar minimum also determines the linear term in optimal logarithmic energy for finitely many points on the sphere.

Corollary 1.3 (Spherical logarithmic energy). *Let $S^2 = \{y \in \mathbb{R}^3 : \|y\| = 1\}$ be the unit sphere. For each integer $n \geq 2$, define*

$$E_{\log}(n) = \min_{(y_1, \dots, y_n) \in (S^2)^n} \left(- \sum_{\substack{1 \leq i, j \leq n \\ i \neq j}} \log \|y_i - y_j\| \right),$$

where the sum is over ordered pairs, $\|\cdot\|$ is the Euclidean norm in $\mathbb{R}^3$, $\log$ is the natural logarithm, and a collision has energy $+\infty$. Then, as $n \rightarrow \infty$ through the integers,

$$E_{\log}(n) = \left(\frac{1}{2} - \log 2 \right) n^2 - \frac{n}{2} \log n + C_{\text{BHS}} n + o(n), \quad (1.7)$$

where

$$C_{\text{BHS}} = 2 \log 2 + \frac{1}{2} \log \left(\frac{2}{3} \right) + 3 \log \left(\frac{\sqrt{\pi}}{\Gamma(1/3)} \right), \quad (1.8)$$

and Γ is Euler's gamma function.

This proves the $d = 2$ case of the logarithmic-energy conjecture of Brauchart, Hardin, and Saff [4, Conjecture 4]. Bétermin and Sandier had already proved the existence of the linear coefficient and shown that it equals (1.8) precisely when the density-one triangular lattice minimizes their planar renormalized energy [1, Theorem 1.5]. Section 8 checks the normalization and applies that equivalence to Theorem 1.1. The conclusion is an asymptotic value, not a construction of near-optimal point sets or an algorithm for Smale's seventh problem.

1.2 Context and prior work

The finite-vortex renormalized energy of Bethuel, Brezis, and Hélein [2] is an antecedent of the infinite-system energy considered here. Sandier and Serfaty introduced the planar functional in their analysis of microscopic vortex patterns in the Ginzburg–Landau model. They proved existence of a minimum, approximation of its value by square-periodic fields, and triangular optimality among Bravais lattices [17, Theorems 1 and 2]. Their Conjecture 1 asks whether the triangular value is also the minimum over the full admissible class. Theorem 1.1 resolves this conjecture positively. The current normalization in that paper has one point per area $2\pi/m$; our density-one convention corresponds to its background $m = 2\pi$.

The lattice comparison belongs to a longer history. The minimization of convergent Epstein zeta sums at fixed covolume was developed by Rankin [14] and Cassels [5], with an error noted in [6] and a repair supplied by Ennola [8] and simplified by Diananda [7]. Montgomery proved triangular minimality of the lattice theta function for every positive Gaussian parameter [11, Theorem 1]. These are comparisons within the lattice class; the issue here is the lower bound for arbitrary admissible configurations.

For the confining-potential class in their Theorem 2, Sandier and Serfaty show that microscopic limits of finite Coulomb minimizers concentrate on minimizers of the planar renormalized energy [18, Theorem 2]. Its minimum also determines a next-order coefficient for optimal logarithmic energy on the sphere [1, Theorem 1.5]. Petrache and Serfaty prove a conditional implication from universal optimality to minimality for Coulomb and Riesz renormalized energies in their configuration-energy normalization [13, Theorem 2 and Corollary 4]. Their periodic comparison uses tori compatible with the reference lattice, not arbitrary square tori, and their planar conclusion is conditional on the Cohn–Kumar conjecture. The companion universal-optimality theorem supplies the Gaussian comparison used in that implication [12, Theorem 1.1 and Corollary 1.3]. The present proof treats the logarithmic functional directly and uses no universal-optimality result as an input.

Two geometric precedents help explain the finite-torus argument. Screening a point charge by a uniform disk yields a separation argument for planar confined logarithmic minimizers; see Lieb, Rougerie, and Yngvason [10, Lemma 3 and Appendix A]. Voronoi moment inequalities combine sharp cell comparisons with Euler’s bound on the average number of sides, as in Gruber’s proof of a theorem of Fejes Tóth [9]. Affine bounds on cell energies, sharp at the regular hexagon, also appear in the Wasserstein model of Bourne, Peletier, and Theil for cells with at least three sides and mass $v \geq 1.5 \cdot 10^{-4}$ [3, Lemma 12]. Here the cell comparison must arise from a trial potential that is continuous off the sites and has the prescribed logarithmic singularities. Its nonradial correction and the two scalar estimates of Section 5 address that requirement, including sectors whose perpendicular foot lies outside the corresponding edge.

The whole-plane reduction below uses Sandier and Serfaty’s local energy estimates and periodic approximation. The finite-torus argument provides the comparison with the triangular value. Throughout, the conclusion concerns energy per unit area: compact changes to a configuration can preserve this value, so the minimum statement alone does not classify all minimizing configurations.

1.3 How the comparison works

The proof separates a finite-torus comparison from the passage to arbitrary fields. The finite comparison identifies the triangular value. Sandier and Serfaty’s approximation theorem supplies periodic fields approaching the unknown full-plane minimum; it does not identify that minimum.

Section 2 checks the conversion from that approximation theorem to our normalization and fixed-width square cutoffs. The main point is to control the boundary error for an arbitrary field whose energy limsup is bounded above, without assuming separation or density one in advance. Local energy estimates give the needed charge density and boundary-strip bound. Proposition 2.4 then transfers any common lower bound on square tori to every field in $\mathcal{A}_1$, including the possible infinite-energy cases.

We next prove the finite comparison. On each square torus the point energy attains a minimum. A screened logarithmic potential and the minimum principle separate the minimizing points, as shown in Section 3. Separation is used only for that minimum; once it is bounded below, every other configuration is bounded below by minimality. This is why the main theorems impose no separation condition on their competitors.

The lifted minimizer has convex Voronoi cells, the regions closer to one site than to any other. Each edge and its incident cell determine a sector joining that edge to the site. Counting incidences on the torus gives at most six sectors per point on average. The comparison is not an estimate of the electrostatic energy of an isolated cell. Instead, we construct a single trial potential H on the torus. Its sector formulas have matching traces and satisfy $H = -\log r + O(r^2)$ near each site. Thus H and the Green potential h have matching logarithmic singularities, while the constant term is normalized to zero in H . Proposition 4.3 gives the exact identity

$$\mathcal{W}_n(h) = \sum_S J(S) + \frac{1}{2} \int_{T_n} |\nabla(h - H)|^2.$$

Here $J(S)$ is the trial potential's sector contribution: its linear background-interaction term minus one half of its renormalized Dirichlet integral, as defined in (4.10). The identity cancels the matching logarithmic singularities and retains the nonnegative square exactly.

The construction of H uses radial functions common to every sector. At a shared Voronoi edge its trace depends only on the equal distances to the two neighboring sites; along a ray to a vertex it depends only on the vertex radius. These observations glue the local formulas continuously off the sites and give an H^1 function on every punctured torus. The identity is proved for any compatible flat torus whose area equals its number of sites, not just for square tori. A separate one-site triangular torus supplies six regular sectors; there the chosen potential has exactly the triangular Green gradient, so the square term vanishes.

Section 5 constructs the radial functions from the triangular Green expansion and subtracts fixed multiples of sector area and angle. The desired bound has the form

$$J(S) \geq K + \lambda A(S) + \mu \theta(S), \quad K < 0.$$

Areas and angles sum exactly to n and $2\pi n$ on a square torus. The incidence count then gives $n(6K + \lambda + 2\pi\mu)$, and exact calibration on the triangular torus identifies this with $nW(E_\Delta)$.

The perpendicular from a site to a Voronoi edge need not lie on the edge, so one endpoint angle of a sector can be negative. A lower bound for a scalar primitive alone does not control the resulting difference of two primitive values. A second estimate bounds the total negative part of its derivative. Together these estimates handle both ordinary and signed sectors without imposing a regular-polygon hypothesis.

Section 6 proves summable analytic bounds for all omitted lattice modes. Section 7 first verifies the retained finite core and then uses the analytic error bounds to enclose the full functions and their derivatives. Midpoint and interpolation errors cover a compact rectangle between mesh points. At the exact triangular point, analytic stationarity and a positive Hessian bound establish the non-strict minimum; an exact formula handles the unbounded exterior. Section 8 combines these results with the geometry and the whole-plane reduction, then derives Corollary 1.3. Appendix A records two complementary methods: an independent coefficient recurrence and sharper exact tail bounds. They preserve additional quantitative information without adding premises to the main comparison.

2 From square tori to the full plane

This section proves the reduction needed for the full admissible class. A uniform lower bound on finite square tori gives the same lower bound for every field in $\mathcal{A}_1$. The argument uses the periodic approximation theorem of Sandier and Serfaty, after checking the normalization and the fixed-width cutoffs in the definition of W .

2.1 Local regularity and the puncture limit

The local integrability convention causes no restriction on the competitors: under the distributional equations, membership in L^1_{loc} already implies membership in every L^q_{loc} , $1 \leq q < 2$.

Lemma 2.1. *Let $\Lambda \subset \mathbb{R}^2$ be simple and locally finite, and let $E \in \mathcal{D}'(\mathbb{R}^2; \mathbb{R}^2)$ be a distributional vector field satisfying*

$$\operatorname{div} E = 2\pi \left(\sum_{p \in \Lambda} \delta_p - 1 \right), \quad \operatorname{curl} E = 0.$$

The field is smooth off Λ . Near each $p \in \Lambda$ it has the form

$$E(x) = \frac{x - p}{|x - p|^2} + F_p(x), \quad F_p \text{ smooth.} \quad (2.1)$$

Consequently E has a representative in L^q_{loc} for every $1 \leq q < 2$, and $W(E, \xi)$ exists as a finite real number for every smooth compactly supported ξ .

Proof. Choose a disk about p containing no other charge and subtract

$$E_p^0(x) = \nabla \log |x - p| - \pi(x - p).$$

Since $\Delta \log |x - p| = 2\pi \delta_p$, the difference $E - E_p^0$ has zero divergence and curl as a distribution on the disk. The identities

$$\Delta V_1 = \partial_1 \operatorname{div} V - \partial_2 \operatorname{curl} V, \quad \Delta V_2 = \partial_2 \operatorname{div} V + \partial_1 \operatorname{curl} V$$

and Weyl's lemma make both components of this difference smooth. On a disk without charges, use $E^0(x) = -\pi x$ instead. This proves (2.1) and smoothness off the charges. The singular function $|x - p|^{-1}$ is locally in L^q exactly when $q < 2$.

Only finitely many charges meet a fixed compact neighborhood of $\operatorname{supp} \xi$. Choose disjoint small disks about them. In each disk,

$$\frac{1}{2} |E(x)|^2 - \frac{1}{2|x - p|^2} \in L^1.$$

Moreover $\xi(x) - \xi(p) = O(|x - p|)$, so $(\xi(x) - \xi(p))/|x - p|^2$ is integrable. The remaining singular integral is $\pi \xi(p) \log(\rho/\eta)$, whose divergence cancels the prescribed $\pi \xi(p) \log \eta$. The integral outside these disks is finite. Thus the puncture limit exists and is finite. $\square$

In particular the only possible infinite values of $W(E)$ arise in the large-square limsup. No puncture limit is interchanged with that limsup below.

2.2 Approximation and local energy estimates

We distinguish the current normalization from the electric-field normalization of $\mathcal{A}_1$. For $b > 0$, write $\mathcal{C}_b$ for currents $j \in L^1_{\text{loc}}(\mathbb{R}^2; \mathbb{R}^2)$ with

$$\operatorname{div} j = 0, \quad \operatorname{curl} j = \rho - b, \quad \rho = 2\pi \sum_{v \in \Lambda} \delta_v, \quad \rho(B_s) = O(s^2),$$

where Λ is simple and locally finite. Denote their local puncture functional by $W(j, \xi)$; its formula is the same as that of $W(E, \xi)$. Put $Q_s = (-s, s)^2$.

For such a current, the inverse component rotation $F = (j_2, -j_1)$ satisfies $\operatorname{div} F = \rho - b$ and $\operatorname{curl} F = 0$. In the proof of Lemma 2.1, use the local model $\nabla \log |x - p| - (b/2)(x - p)$ near a charge and $-(b/2)x$ off the charges. This gives the radial singular expansion for F and its rotated expansion for j . Orthogonality preserves the squared norm, so the same finite local puncture conclusion holds for j and every smooth compactly supported cutoff. This local argument uses simplicity and local finiteness, but not the charge-growth bound.

For balls and for Q_s , choose cutoffs $0 \leq \zeta_{U_s} \leq 1$, compactly supported in U_s , equal to one at distance at least one from its boundary, and with uniformly bounded gradients. Write

$$W_U(j) = \limsup_{s \rightarrow \infty} \frac{W(j, \zeta_{U_s})}{|U_s|}.$$

Theorem 1(1),(3),(4) of [17] makes the cutoff choice immaterial within each fixed shape family, gives a common finite minimum

$$M = \min_{\mathcal{C}_1} W_{\text{ball}} = \min_{\mathcal{C}_1} W_Q \in \mathbb{R}, \quad (2.2)$$

and provides square-periodic $j_k \in \mathcal{C}_1$ with $W_{\text{ball}}(j_k) \rightarrow M$. Their square periods have side lengths $\sqrt{2\pi n_k}$ for positive integers n_k . The common minimum does not assert equality of the ball and square values of an arbitrary fixed field. Their conditions (1.4)–(1.5) require increasing bounded open sets with $\bigcap_{s>0} \overline{U}_s = \{0\}$, left-continuous volumes, $|U_s - U_s| \leq C|U_s|$, and, for every fixed $a \in \mathbb{R}^2$,

$$\frac{|(a + U_s) \triangle U_s|}{|U_s|} \rightarrow 0.$$

They further require constants C and $\theta < 2$ such that

$$U_s + B_1 \subset U_{s+C}, \quad U_{s+1} \subset U_s + B_C, \quad |U_{s+1} \setminus U_s| = O(s^\theta).$$

Balls and squares satisfy these conditions with $\theta = 1$ and $|U_s - U_s| = 4|U_s|$.

We also use the following nonsharp consequences of Lemma 4.7 and Proposition 4.9 of [17]. The latter is an application of the mass-displacement method developed for Ginzburg–Landau energies in [16]. In the proof of their Lemma 4.7, Sandier and Serfaty credit the dyadic-shell argument for the local L^r estimate to Struwe [19]; the precise weighted estimate used here is theirs. Let U be bounded and open, $\hat{U} = U + B_1$, and suppose $\operatorname{div} j = 0$ and $\operatorname{curl} j = \rho - 1$ in $\hat{U}$. The number n of simple charges there is finite, and j is locally square-integrable off those charges. For $1 < r < 2$, $0 \leq \xi \leq 1$ compactly supported in U , and $\|\nabla \xi\|_\infty \leq C_0$, Lemma 4.7 implies

$$\int_U \xi^{r/2} |j|^r \leq C_{r,C_0} (|U| + 1)^{1-r/2} (W(j, \xi)_+ + n \log(2 + n) + 1)^{r/2}, \quad (2.3)$$

where $a_+ = \max\{a, 0\}$. For $n \geq 1$, this follows by replacing the energy by its positive part, enlarging the logarithmic count, and absorbing the fixed cutoff bounds into the constant. The statement of Lemma 4.7 and the final display of its proof differ by a factor two on the energy term; this weakening covers either form. If $n = 0$, there is no puncture term and Hölder's inequality gives directly

$$\int_U \xi^{r/2} |j|^r \leq |U|^{1-r/2} \left(\int_U \xi |j|^2 \right)^{r/2} = |U|^{1-r/2} (2W(j, \xi))^{r/2},$$

which also implies (2.3). Proposition 4.9 provides one measure g_U , supported in $\hat{U}$ and satisfying $g_U \geq -C \, dx$, such that for every smooth compactly supported ξ in U ,

$$\left| W(j, \xi) - \int \xi \, dg_U \right| \leq C n_\xi (\log(2 + n_\xi) + 1) \|\nabla \xi\|_\infty, \quad (2.4)$$

where

$$n_\xi = \#\{v \in \Lambda : B(v, C) \cap \text{supp } \nabla \xi \neq \emptyset\}.$$

The constants in (2.4) are universal. In particular the same g_U works for two different cutoffs. The local smoothness needed in these results follows as in Lemma 2.1, after rotating the components.

2.3 Scaling and comparison of fixed-width cutoffs

Lemma 2.2. *Every $E \in \mathcal{A}_1$ satisfies*

$$W(E) \geq 2\pi \left(M - \frac{1}{4} \log(2\pi) \right). \quad (2.5)$$

In particular $W(E) \neq -\infty$.

Proof. The case $W(E) = +\infty$ is immediate. Assume that its limsup is bounded above, allowing provisionally $W(E) = -\infty$. Set $m = 2\pi$ and rotate the vector components by

$$j = (-E_2, E_1), \quad j'(y) = m^{-1/2} j(y/\sqrt{m}), \quad \xi_t(y) = \chi_{t/\sqrt{m}}(y/\sqrt{m}).$$

Thus $j \in \mathcal{C}_m$ and $j' \in \mathcal{C}_1$; the charges of j' are at $\sqrt{m}\Lambda$. The cutoff ξ_t is supported in $\overline{Q}_t$, equals one on $\overline{Q}_{t-\sqrt{m}}$, and has a uniformly bounded gradient. Changing variables in the punctured integral changes its puncture radius from η to $\sqrt{m}\eta$. Consequently, at each fixed $t = \sqrt{m}R$,

$$W(E, \chi_R) = W(j', \xi_t) - \frac{\pi}{2} \log m \sum_{v \in \sqrt{m}\Lambda} \xi_t(v). \quad (2.6)$$

There is no area factor in this local identity.

The charge-growth hypothesis and the upper bound on the limsup give $W(j', \xi_t) \leq Ct^2$ for all sufficiently large t . For any fixed $A > 0$, apply (2.3) to ξ_{t+D} with a fixed $D > A + \sqrt{m} + 2$, in $U = Q_{t+D+1}$. There are $O(t^2)$ charges in $\widehat{U}$, and this cutoff is one on Q_{t+A} , so

$$\int_{Q_{t+A}} |j'|^r \leq C_{r,A} t^2 (\log t)^{r/2}, \quad 1 < r < 2. \quad (2.7)$$

Here are the charge estimates that make the boundary error negligible. Fix $A > 0$. Coarea for $y \mapsto \|y\|_\infty$ and (2.7) with $A + 2$ in place of A give radii

$$s_+ \in (t + A, t + A + 1), \quad s_- \in (t - A - 1, t - A)$$

such that both boundary integrals of $|j'|^r$ are at most $Ct^2(\log t)^{r/2}$. They may be chosen with no charge on either boundary. Stokes' theorem, first on the punctured square and then letting the punctures shrink, yields

$$\rho'(Q_{s_\pm}) - |Q_{s_\pm}| = \int_{\partial Q_{s_\pm}} j' \cdot \tau \, d\mathcal{H}^1, \quad \rho' = 2\pi \sum_{v \in \sqrt{m}\Lambda} \delta_v.$$

Hölder's inequality bounds the absolute value by $Ct^{1+1/r}(\log t)^{1/2}$. Monotonicity of ρ' therefore gives

$$\rho'(Q_t) = |Q_t| + o(t^2), \quad (2.8)$$

$$\rho'(Q_{t+A} \setminus Q_{t-A}) \leq C_A(t + t^{1+1/r}(\log t)^{1/2}) = o(t^2/\log t). \quad (2.9)$$

The last equality uses $r > 1$. In particular

$$\sum_v \xi_t(v) = \frac{|Q_t|}{2\pi} + o(t^2). \quad (2.10)$$

Choose a fixed $L > \sqrt{m} + 2$ and a standard unit-width cutoff $\zeta_{Q_{t-L}}$. Both this cutoff and ξ_t are compactly supported in $U = Q_{t+1}$. Their gradient supports, together with any fixed-radius neighborhoods, lie in a strip of the form appearing in (2.9). Hence each error in (2.4) is $o(t^2)$. Since $0 \leq \zeta_{Q_{t-L}} \leq \xi_t \leq 1$, and their difference is supported in $\overline{Q_t} \setminus Q_{t-L-1}$,

$$\int (\xi_t - \zeta_{Q_{t-L}}) dg_U \geq -C \int (\xi_t - \zeta_{Q_{t-L}}) dx \geq -O(t).$$

Using this one measure for both cutoffs proves

$$W(j', \xi_t) \geq W(j', \zeta_{Q_{t-L}}) - o(t^2). \quad (2.11)$$

Every boundary width used here is fixed as $t \rightarrow \infty$.

Divide (2.11) by $|Q_t|$ and take limsup. The ratio $|Q_{t-L}|/|Q_t|$ tends to one, so

$$\limsup_{t \rightarrow \infty} \frac{W(j', \xi_t)}{|Q_t|} \geq W_Q(j') \geq M.$$

These inequalities hold in the extended real sense. From (2.6), (2.10), and $|Q_t| = m|K_R|$, we obtain

$$W(E) = m \left(\limsup_{t \rightarrow \infty} \frac{W(j', \xi_t)}{|Q_t|} - \frac{1}{4} \log m \right).$$

This proves (2.5) and rules out the provisional case $W(E) = -\infty$. $\square$

2.4 Periodic averaging and the harmonic component

For a periodic field F with one unit logarithmic singularity at each of N charges in a period cell P , define its cell energy by

$$\mathcal{E}_P(F) = \lim_{\eta \downarrow 0} \left[\frac{1}{2} \int_{P \setminus \bigcup_v B(v, \eta)} |F|^2 dx + \pi N \log \eta \right].$$

The integral is interpreted on the flat torus, so disks meeting a chosen cell boundary are identified periodically.

Lemma 2.3. *Let F be a periodic admissible electric field or current, with simple charges and period cell P of area A . For either square or ball cutoffs with fixed boundary width and uniformly bounded gradients, its energy per unit area has a limit, equal to $\mathcal{E}_P(F)/A$.*

Proof. There are finitely many charges on the fixed torus. Choose $\rho > 0$ smaller than its injectivity radius, with the disks about distinct torus charges disjoint, and write on its periodic lift

$$F_\rho(x) = \frac{1}{2} |F(x)|^2 - \sum_v \frac{\mathbf{1}_{B(v, \rho)}(x)}{2|x - v|^2}.$$

By the local expansion in Lemma 2.1, with rotated components for currents, F_ρ is periodic and belongs to $L^1(P)$. For a cutoff ξ the puncture calculation gives

$$\begin{aligned} W(F, \xi) &= \int \xi F_\rho dx + \pi \log \rho \sum_v \xi(v) + \mathcal{R}_\xi, \\ \mathcal{R}_\xi &= \sum_v \int_{B(v, \rho)} \frac{\xi(x) - \xi(v)}{2|x - v|^2} dx. \end{aligned}$$

Here W denotes the same local formula for either type of field. A summand in $\mathcal{R}_\xi$ vanishes unless $B(v, \rho)$ meets $\text{supp } \nabla \xi$, and each has absolute value at most $\pi \rho \|\nabla \xi\|_\infty$. For a ball or square of size R there are $O(R)$ such charges, with constants allowed to depend on the fixed periodic field. Thus $\mathcal{R}_\xi = O(R)$.

The number of period cells meeting any fixed-width boundary strip is also $O(R)$. Integrating the periodic L^1 function F_ρ over the remaining complete cells, and bounding the boundary cells by $\|F_\rho\|_{L^1(P)}$, shows that

$$\frac{1}{|U_R|} \int \xi F_\rho \, dx \longrightarrow \frac{1}{A} \int_P F_\rho \, dx, \quad \frac{1}{|U_R|} \sum_v \xi(v) \longrightarrow \frac{N}{A}.$$

The resulting limit is $(\int_P F_\rho + \pi N \log \rho)/A = \mathcal{E}_P(F)/A$. $\square$

On $T_n = \mathbb{R}^2/(\sqrt{n}\mathbb{Z}^2)$, let h be the mean-zero solution of $-\Delta h = 2\pi(\sum_{i=1}^n \delta_{v_i} - 1)$ for distinct points $v_1, \dots, v_n$, with energy defined by (1.6). The compatibility condition holds because $|T_n| = n$; the mean-zero solution exists and is unique by the Fourier series for the Laplacian on the torus. Any periodic locally integrable curl-free field E with divergence $-\Delta h$ differs from $-\nabla h$ by a smooth field of zero divergence and curl. Each component is harmonic on the torus, hence constant. Therefore $E = -\nabla h + c$, with $c \in \mathbb{R}^2$. As a distributional derivative of a periodic function, ∇h has integral zero; it is in $L^1(T_n)$ by Lemma 2.1. The cross integral on the punctured torus thus tends to zero, giving the exact identity

$$\mathcal{E}_{T_n}(E) = \mathcal{W}_n(h) + \frac{n}{2}|c|^2. \quad (2.12)$$

Applying the same fixed orthogonal rotation to the fields preserves this energy identity.

Proposition 2.4 (Reduction to square tori). *Let $C \in \mathbb{R}$. Suppose that for every integer $n \geq 2$ and every configuration of n distinct points of T_n , the mean-zero potential in (1.6) satisfies $\mathcal{W}_n(h) \geq nC$. Then $W(E) \geq C$ for every $E \in \mathcal{A}_1$. In particular, if these finite-torus bounds hold with $C = W(E_\Delta)$, then*

$$\inf_{E \in \mathcal{A}_1} W(E) = W(E_\Delta).$$

Proof. Take the square-periodic minimizing sequence $j_k \in \mathcal{C}_1$ in (2.2). Integrating its curl on a square period torus shows that the area of that torus is $2\pi N_k$, where N_k is its positive integer number of charges. If $N_k = 1$, use twice the square period, which has four charges. Lemma 2.3 shows that this change of period preserves the energy per unit area. We may therefore assume $N_k \geq 2$. We may also align each period square with the coordinate axes: simultaneous rotation of the positions and vector components preserves its cell energy, and the same lemma identifies its energy per unit area.

With $m = 2\pi$, rescale and undo the component rotation:

$$E_k(x) = \sqrt{m}((j_k)_2(\sqrt{m}x), -(j_k)_1(\sqrt{m}x)).$$

This is an admissible field on T_{N_k} . Changing the puncture radius in its cell gives

$$\mathcal{E}_{T_{N_k}}(E_k) = \mathcal{E}_{P_k}(j_k) - \frac{\pi}{2}N_k \log m, \quad |P_k| = mN_k.$$

Combining periodic averaging, (2.12), and the assumed torus inequality yields

$$m \left(W_{\text{ball}}(j_k) - \frac{1}{4} \log m \right) = \frac{\mathcal{E}_{T_{N_k}}(E_k)}{N_k} \geq C.$$

Letting $k \rightarrow \infty$ gives $m(M - \frac{1}{4} \log m) \geq C$. Lemma 2.2 now proves the conclusion for every full-plane competitor, including all possible infinite-energy cases.

For the last assertion, the covolume-one triangular lattice is simple, locally finite, and has quadratic charge growth. Its periodic potential gives an element $E_\Delta \in \mathcal{A}_1$, whose finite energy follows from Lemmas 2.1 and 2.3. It supplies the opposite inequality for the infimum. $\square$

3 Finite tori and separated minimizing configurations

We first isolate the finite-dimensional problem used in the periodic reduction. Separation will be proved only for its minimizing configurations. This suffices for a lower bound on every configuration.

For an integer $n \geq 2$, put

$$L = \sqrt{n}, \quad T_n = \mathbb{R}^2 / (L\mathbb{Z}^2).$$

The torus has area n . Given distinct points $v_1, \dots, v_n \in T_n$, let h be the mean-zero distributional solution of

$$-\Delta h = 2\pi \left(\sum_{i=1}^n \delta_{v_i} - 1 \right).$$

We use the energy (1.6). For this square-torus energy, η is smaller than the injectivity radius and half the minimum distance between the finitely many points.

3.1 Green representation and existence

Lemma 3.1 (Green representation and existence). *Let G_n be the mean-zero Green function on T_n , normalized by*

$$-\Delta G_n = 2\pi(\delta_0 - 1/n), \quad R_n = \lim_{x \rightarrow 0} (G_n(x) + \log |x|).$$

For every configuration of n distinct points,

$$h(x) = \sum_{i=1}^n G_n(x - v_i), \quad \mathcal{W}_n(h) = \pi n R_n + \pi \sum_{i \neq j} G_n(v_i - v_j). \quad (3.1)$$

The minimum of $\mathcal{W}_n$ over such configurations is attained at a configuration of distinct points.

Proof. The first identity follows from the equation and the zero-mean normalization. Subtracting the logarithmic singularity and applying local elliptic regularity gives, near v_i ,

$$h(x) = -\log |x - v_i| + a_i + O(|x - v_i|), \quad a_i = R_n + \sum_{j \neq i} G_n(v_i - v_j).$$

Let $\Omega_\eta = T_n \setminus \bigcup_i B(v_i, \eta)$. On its hole boundaries the outward normal points towards v_i ; therefore

$$\partial_n h = \frac{1}{\eta} + O(1).$$

Since $\Delta h = 2\pi$ on Ω_η , integration by parts gives

$$\frac{1}{2} \int_{\Omega_\eta} |\nabla h|^2 = -\pi \int_{\Omega_\eta} h + \frac{1}{2} \sum_i \int_{\partial B(v_i, \eta)} h \partial_n h.$$

The bulk integral tends to zero because h has mean zero. The i -th boundary integral is $2\pi(-\log \eta + a_i) + o(1)$. Taking the puncture limit proves (3.1).

The function G_n is bounded below on its punctured torus and tends to $+\infty$ at the origin. Extend the right-hand side of (3.1) to $+\infty$ on configurations with a collision. This gives a lower semicontinuous function on the compact space T_n^n , finite at every configuration of distinct points. A collision cannot be canceled by other pair terms, since all those terms have a common finite lower bound. A minimum therefore exists and has distinct points. $\square$

By (2.12), every admissible periodic field with these charges differs from $-\nabla h$ by a constant vector c , which adds $n|c|^2/2$ to the energy. It therefore suffices to minimize over the mean-zero potentials considered here.

3.2 Separation of minimizers

The argument uses a screened logarithmic potential and the minimum principle. A closely related exclusion argument for minimizing configurations of a planar confined system, with the same single-charge screening radius, appears in Lieb, Rougerie, and Yngvason [10, Lemma 3 and Appendix A]. The following proof treats the torus Green function and its background directly.

Lemma 3.2 (Separation). *For every $n \geq 2$, a minimizing configuration in Lemma 3.1 has torus distance at least*

$$r_* = \frac{1}{\sqrt{\pi}}$$

between distinct points. Its full lift to $\mathbb{R}^2$ has the same minimum-separation bound, including between different period translates of one torus point.

Proof. Fix one point q of a minimizing configuration. The symmetry of G_n , together with (3.1), shows that

$$\Phi(x) = \sum_{p \neq q} G_n(x - p)$$

has a global minimum at q . At a fixed point $p \neq q$, interpret $\Phi(p) = +\infty$.

The injectivity radius of T_n is $\sqrt{n}/2 > r_*$ for $n \geq 2$. We may therefore define a function on the embedded radius- r_* disk by

$$\phi(x) = \begin{cases} -\log(|x|/r_*) + \frac{\pi}{2}(|x|^2 - r_*^2), & 0 < |x| < r_*, \\ 0, & |x| \geq r_*. \end{cases}$$

Its value and radial derivative vanish at r_* , and it is positive inside the disk. Consequently

$$-\Delta\phi = 2\pi(\delta_0 - \mathbf{1}_{B(0, r_*)})$$

without a boundary measure on the circle. Each such disk has area one.

Set

$$\Omega = \bigcup_{p \neq q} B(p, r_*), \quad \Psi = \Phi - \sum_{p \neq q} \phi(\cdot - p).$$

All poles cancel. In particular $\Psi \in W^{2,s}(T_n)$ for every finite s , and Ψ is continuous. On Ω ,

$$-\Delta\Psi = 2\pi \sum_{p \neq q} \mathbf{1}_{B(p, r_*)} - 2\pi \frac{n-1}{n} \geq \frac{2\pi}{n} > 0.$$

Since $|\Omega| \leq n-1 < n$, every component of Ω has nonempty boundary. Every obstacle vanishes on this boundary, so $\Psi = \Phi$ there.

If $q \in \Omega$, let D be its component. The strict minimum principle, valid for this continuous $W^{2,s}$ supersolution, gives

$$\Phi(q) > \Psi(q) > \min_{\partial D} \Psi = \min_{\partial D} \Phi \geq \Phi(q),$$

a contradiction. No regularity of the union's boundary is required: the minimum is attained on $\overline{D}$, and strict superharmonicity excludes an interior minimum. Hence $q \notin \Omega$. As q was arbitrary, distinct torus points are separated by r_* .

Distances between lifts of distinct torus points are at least their torus distance. Two different lifts of one point differ by a nonzero vector in $L\mathbb{Z}^2$, whose length is at least $L = \sqrt{n} > r_*$. $\square$

4 Voronoi sectors and a glued dual potential

We first record the sector geometry of separated square-torus minimizers. We then prove gluing and duality for compatible periodic Voronoi sectors on any flat torus. This separates the general identity from its square-torus and triangular applications. The construction uses general radial functions; their particular choice and triangular calibration will be given later. Its purpose is to turn a sector lower bound, established by two scalar estimates, into an energy bound for a whole configuration.

The use of polygon inequalities together with an average of at most six sides is familiar from moment methods; see Gruber's proof of Fejes Tóth's theorem [9]. A related affine cell bound for a Wasserstein interaction is sharp at the regular hexagon of unit mass; the result of Bourne, Peletier, and Theil [3, arXiv v3, Lemma 12] assumes integer side counts at least three and cell mass $v \geq 1.5 \cdot 10^{-4}$. Here we must construct a globally admissible singular potential before estimating its sector contributions. The arguments below supply that construction and the incidence count on a torus.

4.1 Counting sectors on the torus

For $d > 0$ and real α, β with $|\alpha|, |\beta| < \pi/2$ and $\alpha + \beta > 0$, define the sector

$$S(d, \alpha, \beta) = \{(r, \varphi) : -\alpha < \varphi < \beta, 0 < r < d \sec \varphi\}. \quad (4.1)$$

Its polar axis is the normal from its site toward the supporting edge.

Lemma 4.1 (Periodic Voronoi geometry). *Let $P \subset \mathbb{R}^2$ be the full $L\mathbb{Z}^2$ -periodic lift of n distinct points on T_n , and suppose P has separation at least r_* . Its Voronoi polygons, modulo $L\mathbb{Z}^2$, have at most $6n$ face-edge incidences.*

Each such incidence determines a sector (4.1) in polar coordinates about its site, with $d \geq r_/2$. The sectors in one torus have total area n and total angle $2\pi n$. Either endpoint parameter α or β is allowed to be negative.*

Proof. For $p \in P$, the Voronoi cell

$$C_p = \{x : |x - p| \leq |x - q| \text{ for all } q \in P\}$$

contains the disk of radius $r_*/2$ about p . Comparison with the four sites $p \pm (L, 0)$ and $p \pm (0, L)$ gives

$$C_p \subset p + [-L/2, L/2]^2.$$

Local finiteness now implies that C_p is a bounded convex polygon with finitely many sides. There are finitely many polygon, edge, and vertex orbits under the period lattice.

Use maximal positive-length Voronoi edges. Each edge has two face incidences, and each vertex has degree at least three. Indeed a Voronoi vertex has at least three nearest sites; a degenerate vertex with more nearest sites only increases the degree. The quotient is a finite cell decomposition of the torus with $F = n$ open faces. Each open face is a disk: distinct translates of the interior of a lift cell are disjoint. A projected closed face need not be embedded, since its boundary may identify to itself. These identifications are permitted in the cell decomposition.

Let V and E denote its numbers of vertices and edges. Euler's identity and the degree count give

$$V - E + n = 0, \quad 3V \leq 2E.$$

Thus $E \leq 3n$, and the number of face-edge incidences is $M_0 = 2E \leq 6n$. These counts include every occurrence in a face boundary: an edge may be a loop, a face may lie on both

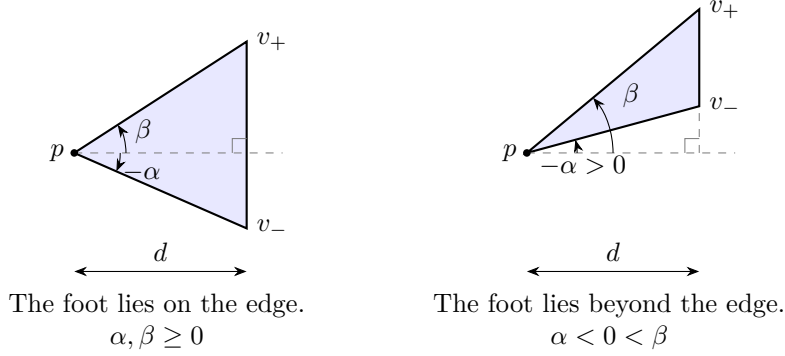


Figure 1: Two possible Voronoi sectors, with the edge normal as angular axis. In the right-hand sector both endpoints are above that axis, so the lower endpoint parameter α is negative even though the sector angle $\alpha + \beta$ is positive.

sides of an edge, a neighbor may recur, and a vertex may occur several times in one boundary. No generic-position assumption is used.

For an edge of C_p shared with C_q , choose as polar axis the unit vector from p towards q . Its supporting line has distance $d = |p - q|/2 \geq r_*/2$ from p . The angular coordinates of its two endpoints satisfy $-\pi/2 < \varphi_- < \varphi_+ < \pi/2$. Write $\varphi_- = -\alpha$ and $\varphi_+ = \beta$. The triangle from p to the edge is exactly (4.1). If the perpendicular foot is outside the edge, one of α, β is negative.

The sectors partition each polygon up to their boundary rays. A polygon has angle 2π about its interior site. Summing areas and angles over the n quotient faces proves the last assertions. $\square$

For reference, the area and angle of (4.1) are

$$A(S) = \frac{1}{2} \int_{-\alpha}^{\beta} d^2 \sec^2 \varphi \, d\varphi, \quad \theta(S) = \alpha + \beta. \quad (4.2)$$

In particular the sector has positive angle even when one endpoint parameter is negative.

4.2 Compatible flat-torus data and trace matching

Let $\Gamma \subset \mathbb{R}^2$ be a rank-two lattice and put $\mathbb{T} = \mathbb{R}^2/\Gamma$. Let $N \geq 1$ be an integer, let $v_1, \dots, v_N$ be distinct points of $\mathbb{T}$, and let P be their full Γ -periodic lift. Suppose that the Voronoi cells of P are bounded convex polygons with finitely many face-edge incidences modulo Γ . Assume that each incidence has the sector form (4.1) and that, for one named constant $d_0 > 0$, every sector has $d \geq d_0$. We call these *compatible periodic Voronoi data with lower edge distance d_0* . This is conditional geometric data, not a separation assertion about an arbitrary competitor.

The sectors partition the N cells modulo Γ , up to their boundary rays. Their total area is $|\mathbb{T}|$, and their total angle is $2\pi N$: each convex cell has its site in its interior because $d_0 > 0$, so its sectors have total angle 2π . These facts do not require an incidence bound.

Call a radius $\eta > 0$ admissible if it is smaller than the injectivity radius of $\mathbb{T}$ and, when $N \geq 2$, smaller than half the minimum torus distance between distinct sites. Thus the torus disks $B(v_i, \eta)$ are embedded and pairwise disjoint; for $N = 1$ there is no pairwise-distance condition. Write

$$\Omega_\eta = \mathbb{T} \setminus \bigcup_{i=1}^N B(v_i, \eta).$$

In puncture limits we may further take $\eta < d_0$, so every deleted disk lies in its Voronoi cell. This restriction does not change a limit as $\eta \downarrow 0$.

Fix compatible periodic Voronoi data with lower edge distance d_0 . Let I be a finite or countable index set, and choose integers $k_i \geq 6$, $i \in I$; different indices may have the same integer. Suppose that

$$B \in C^1((0, \infty)), \quad D_i \in C^1((0, \infty))$$

satisfy the following assumptions:

$$B(r) = -\log r + O(r^2), \quad B'(r) = -1/r + O(r) \quad (r \downarrow 0), \quad (4.3)$$

and, for every $T_0 > d_0$,

$$\sum_{i \in I} \sup_{d_0 \leq t \leq T_0} ((1 + k_i)|D_i(t)| + t|D'_i(t)|) < \infty. \quad (4.4)$$

These are the exact inputs needed in the gluing and duality argument. The later radial construction will verify them directly.

For a sector (4.1), define $t(\varphi) = d \sec \varphi$ and

$$H(r, \varphi) = B(r) + \sum_{i \in I} \left(\frac{r}{t(\varphi)} \right)^{k_i} D_i(t(\varphi)). \quad (4.5)$$

The same radial functions are used in every sector.

Lemma 4.2 (Gluing and singular normalization). *For compatible periodic Voronoi data on $\mathbb{T}$ with $N \geq 1$ distinct sites and lower edge distance d_0 , let I be finite or countable, let $k_i \geq 6$ be integers with repetitions permitted, and let $B, D_i \in C^1((0, \infty))$ satisfy (4.3)–(4.4). The sector functions (4.5) define a Γ -periodic function H that is continuous off the sites and belongs to $H^1(\Omega_\eta)$ for every admissible $\eta < d_0$. At every site, uniformly in the angular variable,*

$$H = -\log r + O(r^2), \quad \nabla H = -\frac{e_r}{r} + O(r). \quad (4.6)$$

The gradient statement is sectorwise, and hence holds for the weak gradient almost everywhere; no agreement of derivatives across sectors is asserted. In particular the constant term in $H + \log r$ is zero.

Proof. Fix the finite collection of lifted polygons needed for one torus. Their vertex radii are bounded by some $T_0 > d_0$. On each closed sector, $d_0 \leq t \leq T_0$, $r/t \leq 1$, and $\tan \varphi$ is bounded. The derivatives of a summand are

$$\partial_r \left[(r/t)^{k_i} D_i(t) \right] = \frac{k_i}{r} (r/t)^{k_i} D_i(t),$$

and

$$\partial_\varphi \left[(r/t)^{k_i} D_i(t) \right] = \tan \varphi (r/t)^{k_i} (t D'_i(t) - k_i D_i(t)).$$

Assumption (4.4) therefore gives uniform convergence of the series and its first derivatives on each closed sector away from its site. Derivatives here are sectorwise; derivatives across different sectors need not agree.

On a polygon edge, $r = t$, so the trace is

$$B(r) + \sum_i D_i(r).$$

The two adjacent sites have the same distance to that edge point, and hence give the same trace. On a ray from a site to a vertex at distance r_v , either neighboring sector has $t = r_v$ at its endpoint angle. Both traces along the entire ray are

$$B(r) + \sum_i (r/r_v)^{k_i} D_i(r_v).$$

At a multiple vertex all incident sites have the same vertex distance, so the edge traces also coincide there. The recipe is unchanged by period translations.

Thus all traces match. Applying the Sobolev gluing property to the finitely many sectors gives the asserted H^1 regularity off the holes. No matching of normal derivatives is required.

For $r \leq d_0/2$, the fact that every $k_i \geq 6$, together with (4.4), bounds the correction series by $O(r^6)$ and its gradient by $O(r^5)$. Combining these bounds with (4.3) proves (4.6). $\square$

The common notation $D_i = b_i - u_i$, $B = \sum_i u_i$ is one way to supply these data. In that notation the common edge trace is simply $\sum_i b_i(r)$. The gluing argument itself only uses B and D_i .

4.3 The exact dual identity

Assume now that $|\mathbb{T}| = N$. Let h be the mean-zero distributional solution on $\mathbb{T}$ of

$$-\Delta h = 2\pi \left(\sum_{i=1}^N \delta_{v_i} - 1 \right). \quad (4.7)$$

The area condition is exactly the compatibility condition for this equation. With the convention

$$\Gamma^* = \{k \in \mathbb{R}^2 : k \cdot \gamma \in \mathbb{Z} \text{ for every } \gamma \in \Gamma\},$$

the Fourier mode $e^{2\pi i k \cdot x}$ has eigenvalue $4\pi^2 |k|^2$ under $-\Delta$. The constant Fourier coefficient of the right-hand side vanishes. Dividing its coefficient at each nonzero $k \in \Gamma^*$ by $4\pi^2 |k|^2$, and setting the coefficient at zero to zero, gives a distributional solution. It is unique because the kernel of the Laplacian on the torus consists of constants. Define

$$\mathcal{W}_{\mathbb{T}}(h) = \lim_{\eta \downarrow 0} \left[\frac{1}{2} \int_{\Omega_\eta} |\nabla h|^2 \, dx + \pi N \log \eta \right], \quad (4.8)$$

where the limit is taken through admissible radii. The local calculation in Lemma 2.1, applied in a torus chart, gives the finite puncture limit. When $\mathbb{T} = T_n$ and $N = n$, this is exactly $\mathcal{W}_n(h)$ in (1.6).

For a sector $S = S(d, \alpha, \beta)$, define

$$\int_S^{\text{ren}} |\nabla H|^2 = \lim_{\eta \downarrow 0} \left[\int_{S \cap \{r > \eta\}} |\nabla H|^2 \, dx + \theta(S) \log \eta \right]. \quad (4.9)$$

Lemma 4.2 shows that this limit is finite. Its counterterm uses the positive angle $\theta(S)$, also for a sector with a negative endpoint parameter. Define

$$J(S) = -2\pi \int_S H \, dx - \frac{1}{2} \int_S^{\text{ren}} |\nabla H|^2. \quad (4.10)$$

The sectorwise estimates in the proof of Lemma 4.2 also show that these expressions are well defined on any individual sector with $d \geq d_0$ and the endpoint conditions in (4.1), without first choosing a periodic tiling.

Proposition 4.3 (Dual identity). *For compatible periodic Voronoi data on $\mathbb{T} = \mathbb{R}^2/\Gamma$ with $N \geq 1$ distinct sites, lower edge distance $d_0 > 0$, and $|\mathbb{T}| = N$, let h be the mean-zero potential (4.7). Let I be finite or countable, let $k_i \geq 6$ be integers with repetitions permitted, and let $B, D_i \in C^1((0, \infty))$ satisfy (4.3)–(4.4). Let H be the function supplied by Lemma 4.2. Then*

$$\mathcal{W}_{\mathbb{T}}(h) = \sum_S J(S) + \frac{1}{2} \int_{\mathbb{T}} |\nabla(h - H)|^2 dx \geq \sum_S J(S). \quad (4.11)$$

The sum runs over all face–edge incidences in one torus. No incidence bound or square-torus separation hypothesis is imposed.

Proof. Apply Lemma 2.1 in a local chart to $-\nabla h$. Near each site, $h = -\log r + a_i + O(r)$ and $\nabla h = -e_r/r + O(1)$. Thus both h and H have singular gradient $-e_r/r$. Their difference has square-integrable gradient. Also (4.6) implies that

$$I_{\text{ren}}(H) = \lim_{\eta \downarrow 0} \left[\int_{\Omega_\eta} |\nabla H|^2 + 2\pi N \log \eta \right]$$

exists. For admissible $\eta < d_0$, the punctured sectors partition Ω_η up to their boundaries. The sector angles sum to $2\pi N$, so $I_{\text{ren}}(H)$ equals the sum of (4.9).

Green’s identity, with h smooth on the punctured torus and $H \in H^1(\Omega_\eta)$, gives

$$\int_{\Omega_\eta} \nabla h \cdot \nabla H = -2\pi \int_{\Omega_\eta} H + \sum_i \int_{\partial B(v_i, \eta)} H \partial_n h.$$

The outward hole normal gives $\partial_n h = 1/\eta + O(1)$. The zero constant term in (4.6) therefore gives

$$\int_{\Omega_\eta} \nabla h \cdot \nabla H = -2\pi \int_{\mathbb{T}} H - 2\pi N \log \eta + o(1).$$

Expanding $|\nabla(h - H)|^2$, inserting this identity, and taking the puncture limit proves

$$\mathcal{W}_{\mathbb{T}}(h) = -2\pi \int_{\mathbb{T}} H - \frac{1}{2} I_{\text{ren}}(H) + \frac{1}{2} \int_{\mathbb{T}} |\nabla(h - H)|^2.$$

Partitioning the first two terms into sectors proves (4.11). $\square$

If the gradient hypotheses were unchanged but the value expansion were instead $H = -\log r + c_i + o(1)$ at v_i , the right-hand side of the last identity would contain the additional term $2\pi \sum_i c_i$. This explains why the zero constant term in the radial construction is part of the normalization.

For a square-torus minimizer, Lemma 4.1 supplies the compatible data with $\Gamma = \sqrt{n} \mathbb{Z}^2$, $N = n$, and $d_0 = d_* = r_*/2 = 1/(2\sqrt{\pi})$. Its torus has area n , so Proposition 4.3 applies with $\mathcal{W}_{\mathbb{T}} = \mathcal{W}_n$. The triangular application in the next section supplies its own period and sectors and does not use a square-torus minimizer.

4.4 From scalar estimates to configuration bounds

The two scalar bounds below control ordinary and signed sectors by different arguments. Once they yield an affine lower bound for each sector, the dual identity and the incidence count give the energy bound. We formulate this implication for compatible flat tori first; separation of a square-torus minimizer then removes the geometric hypotheses from the finite-volume comparison.

Lemma 4.4 (Signed endpoint parameters). *Let $K < 0$ and $d_0 > 0$. For each $d \geq d_0$, let $G(d, \cdot)$ be an even locally integrable real function on $\mathbb{R}$, and put*

$$g_d(x) = \int_0^x G(d, s) \, ds.$$

Suppose that, for every $d \geq d_0$,

$$2g_d(x) \geq K \quad (x \geq 0), \quad \int_0^\infty \max\{0, -G(d, s)\} \, ds \leq -K. \quad (4.12)$$

Then, for every $d \geq d_0$ and every pair α, β with $|\alpha|, |\beta| < \pi/2$ and $\alpha + \beta > 0$,

$$g_d(d \tan \alpha) + g_d(d \tan \beta) \geq K. \quad (4.13)$$

Proof. Evenness of $G(d, \cdot)$ makes g_d odd. If $\alpha, \beta \geq 0$, the first inequality in (4.12) bounds each term in (4.13) below by $K/2$.

If $\alpha < 0$, set

$$u = -d \tan \alpha, \quad v = d \tan \beta.$$

The inequality $\alpha + \beta > 0$ gives $\beta > -\alpha > 0$. Since the tangent is increasing on $(-\pi/2, \pi/2)$, we have $0 < u < v$. Oddness and the second inequality in (4.12) give

$$\begin{aligned} g_d(d \tan \alpha) + g_d(d \tan \beta) &= g_d(v) - g_d(u) = \int_u^v G(d, s) \, ds \\ &\geq - \int_0^\infty \max\{0, -G(d, s)\} \, ds \geq K. \end{aligned}$$

The case $\beta < 0$ is symmetric. Both endpoints cannot be negative because their sum is positive, so these cases are exhaustive. $\square$

Proposition 4.5 (From sector bounds to energy bounds). *Let $d_0 > 0$, let I be finite or countable, let $k_i \geq 6$ be integers with repetitions permitted, and let $B, D_i \in C^1((0, \infty))$ satisfy (4.3) and (4.4) with this lower distance d_0 . For every sector (4.1) with $d \geq d_0$, use these same radial data in (4.5) and let $J(S)$ be the functional (4.10). Suppose that $K \leq 0$ and $\lambda, \mu \in \mathbb{R}$ satisfy*

$$\begin{aligned} J(S(d, \alpha, \beta)) &\geq K + \lambda A(S(d, \alpha, \beta)) \\ &\quad + \mu \theta(S(d, \alpha, \beta)), \end{aligned} \quad (4.14)$$

for every $d \geq d_0$, $|\alpha|, |\beta| < \pi/2$, and $\alpha + \beta > 0$.

Compatible tori. Let $\mathbb{T} = \mathbb{R}^2/\Gamma$ have area N , where $N \geq 1$ is an integer. Suppose N distinct sites supply compatible periodic Voronoi data with lower edge distance d_0 and at most $6N$ face-edge sectors. Their mean-zero potential (4.7) satisfies

$$\mathcal{W}_{\mathbb{T}}(h) \geq N(6K + \lambda + 2\pi\mu). \quad (4.15)$$

All square-torus configurations. Assume in addition $d_0 \leq d_* = 1/(2\sqrt{\pi})$. For every integer $n \geq 2$ and every configuration of n distinct points $v_1, \dots, v_n$ on $T_n = \mathbb{R}^2/(\sqrt{n}\mathbb{Z}^2)$, let h be the mean-zero solution of

$$-\Delta h = 2\pi \left(\sum_{i=1}^n \delta_{v_i} - 1 \right).$$

Then its puncture-first energy (1.6), with embedded and pairwise disjoint punctures, satisfies

$$\mathcal{W}_n(h) \geq n(6K + \lambda + 2\pi\mu). \quad (4.16)$$

More generally, if a periodic locally integrable field E on T_n satisfies, distributionally,

$$\operatorname{div} E = 2\pi \left(\sum_{i=1}^n \delta_{v_i} - 1 \right), \quad \operatorname{curl} E = 0,$$

then $E = -\nabla h + c$ for a constant $c \in \mathbb{R}^2$, and its energy with the same punctures and counterterm obeys

$$\mathcal{E}_{T_n}(E) = \mathcal{W}_n(h) + \frac{n}{2}|c|^2 \geq n(6K + \lambda + 2\pi\mu).$$

The energy identity remains valid after applying the same fixed orthogonal rotation to both fields.

Proof. For compatible data, the sectors have total area N and total angle $2\pi N$. Write $M_0 \leq 6N$ for their number. Proposition 4.3 and (4.14) give

$$\begin{aligned} \mathcal{W}_{\mathbb{T}}(h) &\geq \sum_S J(S) \\ &\geq M_0 K + \lambda N + 2\pi\mu N \\ &\geq N(6K + \lambda + 2\pi\mu). \end{aligned}$$

The last inequality uses $M_0 \leq 6N$ and $K \leq 0$; it includes $K = 0$, when both terms involving K vanish.

Now assume $d_0 \leq d_*$ and fix $n \geq 2$. Lemma 3.1 supplies a minimizing configuration of distinct points, with potential $h_{\min}$. Lemma 3.2 separates its full periodic lift by $r_* = 1/\sqrt{\pi}$. Lemma 4.1 then gives at most $6n$ sectors, each with edge distance at least $d_* \geq d_0$. These are compatible data on T_n , so (4.15) applies to $h_{\min}$. Every other distinct configuration has energy at least $\mathcal{W}_n(h_{\min})$, proving (4.16) without a separation assumption on that configuration.

Equation (2.12) applies to the displayed divergence and curl conditions and gives the constant harmonic component and its exact energy penalty. A fixed orthogonal rotation preserves the squared norm and the puncture counterterm, giving the last assertion. $\square$

The next section chooses particular radial data and exact constants $K < 0, \lambda, \mu$. For these data, the primitive identity (5.15) and the two scalar bounds yield

$$J(S) \geq K + \lambda A(S) + \mu \theta(S)$$

for every relevant sector. Proposition 5.4 applies the compatible-torus conclusion to this construction and identifies $6K + \lambda + 2\pi\mu$ with the triangular energy. Section 8 uses the square-torus conclusion, whose minimizer argument has already removed the separation hypothesis.

5 Calibration on the triangular cell

We construct the radial functions used by the Voronoi test potential. The potential has the same gradient as the triangular Green function on a regular hexagon, while its energy on every other sector reduces to two scalar estimates. The continuation outside that hexagon is auxiliary; its explicit form makes those estimates accessible to interval arithmetic.

5.1 The local triangular Green function

Set

$$a = \sqrt{\frac{2}{\sqrt{3}}}, \quad p = \frac{a}{2}, \quad R_0 = \frac{2p}{\sqrt{3}}, \quad x_0 = \frac{R_0}{2}. \quad (5.1)$$

Thus p and R_0 are the inradius and circumradius of the covolume-one triangular Voronoi hexagon, and $x_0 = p \tan(\pi/6)$. Identify the plane with $\mathbb{C}$, and write $\partial_z = (\partial_x - i\partial_y)/2$, with a nearest lattice point on the positive real axis. For $m \geq 1$ put

$$k_m = 6m, \quad c_m = \frac{1}{k_m} \sum_{z \in \Lambda_\Delta \setminus \{0\}} z^{-k_m}. \quad (5.2)$$

These sums converge absolutely and are real by reflection symmetry. We also introduce an auxiliary index 0, with $k_0 = 6$. It is distinct from index 1: both exponents equal 6, but they will carry different radial functions.

Lemma 5.1. *The mean-zero triangular Green function has the expansion*

$$h_\Delta(z) = -\log|z| + C_\Delta + \frac{\pi|z|^2}{2} + \sum_{m \geq 1} c_m \operatorname{Re}(z^{6m}), \quad 0 < |z| < a, \quad (5.3)$$

where C_Δ is its regular part at the origin. The series and all of its derivatives converge locally away from $|z| = a$.

Proof. Away from the lattice, $\Delta h_\Delta = 2\pi$, so $2\partial_z^2 h_\Delta$ is a meromorphic elliptic function. Its double pole at each lattice point has principal part $(z - \lambda)^{-2}$. Subtract the absolutely locally convergent Weierstrass series

$$\wp(z) = \frac{1}{z^2} + \sum_{\lambda \neq 0} ((z - \lambda)^{-2} - \lambda^{-2}).$$

The difference is holomorphic and periodic, hence constant. If $\omega = e^{\pi i/3}$, invariance of h_Δ under $z \mapsto \omega z$ implies that its second complex derivative transforms by ω^{-2} ; the same is true of $\wp$. Their constant difference must therefore vanish. Expanding each summand for $|z| < a$ gives

$$\wp(z) = z^{-2} + \sum_{j \geq 1} (j+1) z^j \sum_{\lambda \neq 0} \lambda^{-j-2}.$$

Rotation makes the inner sum zero unless $j+2$ is a multiple of 6. Twice complex differentiation of the right side of (5.3) gives this series, because $k_m(k_m - 1)c_m = (k_m - 1) \sum \lambda^{-k_m}$. The difference is a real harmonic affine function after subtracting $\pi|z|^2/2$; rotation removes its linear part and leaves C_Δ . Normal convergence follows either from this power series or from the absolutely convergent lattice sums on every smaller disk. $\square$

5.2 Matching the regular hexagon

Write $\gamma(r) = -\log r + \pi r^2/2$ for $r > 0$. We first choose the radial data on the regular hexagon, where the test potential must equal $h_\Delta - C_\Delta$. In a sector whose outward normal is the positive real axis, the edge has distance p and the endpoint angles are $\pm\pi/6$. At an edge point of radius $t = p \sec \varphi$,

$$p + i\sqrt{t^2 - p^2} = te^{i|\varphi|}.$$

The boundary value of the m th Green-function mode is therefore $c_m t^{k_m} \cos(k_m \varphi)$. Expressing it as a function of the radius gives the polynomial

$$V_m(r) = c_m \sum_{j=0}^{k_m/2} (-1)^j \binom{k_m}{2j} p^{k_m-2j} (r^2 - p^2)^j, \quad p \leq r \leq R_0. \quad (5.4)$$

On either vertex ray, the same mode equals $(-1)^m c_m r^{k_m}$, since $k_m \pi/6 = m\pi$. This is the radial baseline we use for that mode inside the hexagon. The correction in (4.5) then recovers the angular dependence exactly:

$$(-1)^m c_m r^{k_m} + (r/t)^{k_m} (V_m(t) - (-1)^m c_m t^{k_m}) = c_m r^{k_m} \cos(k_m \varphi).$$

Thus the edge data $b_m = V_m$ and the vertex-ray baseline determine the desired potential on each regular sector. Its remaining part γ is already radial and needs no correction there.

5.3 Continuation to arbitrary sectors

Other Voronoi sectors can have edge radii below p or above R_0 . We now extend the radial data to those radii, keeping two derivatives and making the correction vanish beyond a fixed radius. The assembled potential is required to agree with $h_\Delta - C_\Delta$ only inside the regular hexagon; the continuation is chosen for the scalar estimates, not as an extension of that Green function.

Define the clipped polynomial

$$S(y) = \begin{cases} 0, & y \leq 0, \\ 10y^3 - 15y^4 + 6y^5, & 0 < y < 1, \\ 1, & y \geq 1. \end{cases}$$

Let $h_s = 3/40$ and $c = 3/50$. For $0 \leq z \leq h_s$ put

$$P(z) = z - \frac{6z^3}{h_s^2} + \frac{8z^4}{h_s^3} - \frac{3z^5}{h_s^4}, \quad Q(z) = \frac{z^2}{2} - \frac{3z^3}{2h_s} + \frac{3z^4}{2h_s^2} - \frac{z^5}{2h_s^3}.$$

For a function V with two left derivatives at R_0 , its continuation is defined for $r \geq R_0$ by

$$(\mathcal{H}V)(r) = \begin{cases} V(R_0) + P(r - R_0)V'(R_0) + Q(r - R_0)V''(R_0), & r \leq R_0 + h_s, \\ V(R_0), & r > R_0 + h_s. \end{cases} \quad (5.5)$$

Indeed $(P, P', P'')(0) = (0, 1, 0)$ and $(Q, Q', Q'')(0) = (0, 0, 1)$; all six values at h_s vanish. Thus this operation preserves two derivatives at R_0 and joins a constant with two derivatives at $R_0 + h_s$.

Define $b_0 = \gamma$ up to R_0 , continued by (5.5). For $m \geq 1$, set $b_m = V_m$ on $[p, R_0]$ and use $\mathcal{H}V_m$ above R_0 . Below p put

$$b_m(r) = \begin{cases} 0, & 0 < r \leq p - c, \\ S((r - p + c)/c)(T_{0m} + T_{1m}(r - p) + T_{2m}(r - p)^2), & p - c < r < p, \end{cases} \quad (5.6)$$

where

$$\begin{aligned} T_{0m} &= c_m p^{k_m}, \\ T_{1m} &= -\frac{k_m(k_m - 1)}{p} T_{0m}, \\ T_{2m} &= \frac{k_m(k_m - 1)}{p^2} \left(\frac{(k_m - 2)(k_m - 3)}{6} - \frac{1}{2} \right) T_{0m}. \end{aligned}$$

Direct differentiation of (5.4) gives $T_{0m} = V_m(p)$, $T_{1m} = V'_m(p)$, and $2T_{2m} = V''_m(p)$. Consequently every b_i is C^2 away from the origin.

Now define

$$U_m(r) = (-1)^m c_m r^{k_m} (1 - S((r - R_0)/(1/10))), \quad \tau(r) = S\left(\frac{r - R_0 - 1/1000}{3/40}\right),$$

and set

$$\begin{aligned} u_0 &= (1 - \tau)\gamma + \tau b_0, & u_m &= (1 - \tau)U_m + \tau b_m \quad (m \geq 1), \\ B &= \sum_{i \geq 0} u_i, & D_i &= b_i - u_i, & l &= R_0 + \frac{19}{250}. \end{aligned} \quad (5.7)$$

Here the auxiliary mode 0 lets the radial part γ join its constant continuation by the same correction mechanism as the other modes. Its exponent $k_0 = 6$ makes this correction vanish at the pole without changing the singular normalization. The coefficient estimates proved below justify the sums and their first three derivatives on every open spline interval. In particular, B, D_i are C^2 and piecewise C^3 away from zero. Near zero, $B = \gamma + \sum_{m \geq 1} (-1)^m c_m r^{k_m}$; normal convergence of (5.3) therefore gives

$$\begin{aligned} B(r) &= -\log r + O(r^2), & B'(r) &= -1/r + O(r) \quad (r \downarrow 0), \\ D_i(r) &= 0, & B(r) &= B(l) \quad (r \geq l). \end{aligned} \quad (5.8)$$

The fixed radial data also satisfy (4.4) for every $d_0 > 0$. Indeed, there are only finitely many retained modes, while for the omitted modes (6.4) and the vanishing of D_i beyond l bound the summand on $[d_0, T_0]$ by $(1 + k_i + T_0)w_{k_i}$. The explicit geometric terms in (6.3) make this series summable. This conclusion uses the analytic mode bounds, not the later finite-core estimates.

For a sector with edge distance d and endpoint angles $-\alpha < \beta$, write $t(\varphi) = d \sec \varphi$ and use the test potential

$$H(r, \varphi) = B(r) + \sum_{i \geq 0} \left(\frac{r}{t(\varphi)}\right)^{k_i} D_i(t(\varphi)), \quad 0 < r < t(\varphi), \quad -\alpha < \varphi < \beta. \quad (5.9)$$

At the edge its trace is $\sum_i b_i(t)$; at a ray ending at a vertex of radius r_v its trace is $B(r) + \sum_i (r/r_v)^{k_i} D_i(r_v)$. These expressions explain both trace matching conditions in the glued Voronoi construction.

5.4 The sector integral

The following formulas separate radial integration from sector geometry. For $t > 0$ define

$$\begin{aligned} B_1(t) &= B'(t) + 1/t, & S_1(t) &= \int_0^t B(r) r \, dr, \\ S_2(t) &= \int_0^t (-2B_1(r) + r B_1(r)^2) \, dr, \\ H_k(t) &= t^{-k} \int_0^t B_1(r) r^k \, dr, & Q_i(t) &= t D'_i(t) - k_i D_i(t). \end{aligned} \quad (5.10)$$

Here H_k is indexed by the exponent, so the two modes with exponent six use the same H_6 . They remain separate in every sum involving D_i . Put

$$\begin{aligned}
f_0(t) = & -2\pi \left(S_1(t) + t^2 \sum_i \frac{D_i(t)}{k_i + 2} \right) \\
& - \frac{1}{2} \left(\log t + S_2(t) + \sum_i (-2D_i(t) + 2k_i D_i(t) H_{k_i}(t)) \right. \\
& \left. + \sum_{i,j} \frac{k_i k_j}{k_i + k_j} D_i(t) D_j(t) \right), \\
e(t) = & \frac{1}{2} \sum_{i,j} \frac{Q_i(t) Q_j(t)}{k_i + k_j}.
\end{aligned} \tag{5.11}$$

All sums here and below include index 0. The angular coefficient is nonnegative, since

$$e(t) = \frac{1}{2} \int_0^1 \left(\sum_i Q_i(t) v^{k_i} \right)^2 \frac{dv}{v} \geq 0. \tag{5.12}$$

Let J denote the sector contribution to the dual energy,

$$J = -2\pi \int_{\text{sector}} H - \frac{1}{2} \int_{\text{sector}}^{\text{ren}} |\nabla H|^2,$$

where the renormalized integral adds $(\alpha + \beta) \log \eta$ before letting the radial exclusion η tend to zero. Then

$$J = \int_{-\alpha}^{\beta} (f_0(d \sec \varphi) - \tan^2 \varphi e(d \sec \varphi)) d\varphi. \tag{5.13}$$

To verify this identity, differentiate (5.9):

$$H_r = -\frac{1}{r} + B_1(r) + \sum_i \frac{k_i}{r} (r/t)^{k_i} D_i(t), \quad H_\varphi = \tan \varphi \sum_i (r/t)^{k_i} Q_i(t).$$

The source integral gives $S_1 + t^2 \sum_i D_i/(k_i + 2)$. The radial square contributes $\log t + S_2$, its cross terms contribute $-2D_i + 2k_i D_i H_{k_i}$, and its quadratic terms have coefficient $k_i k_j/(k_i + k_j)$. Integrating the angular square gives $\tan^2 \varphi \sum Q_i Q_j/(k_i + k_j)$. These are precisely (5.11). The estimates below permit termwise integration.

5.5 Subtracting area and angle

Given real constants λ, μ , define

$$\begin{aligned}
f(t) &= f_0(t) - \lambda t^2/2 - \mu, & t &= \sqrt{d^2 + s^2}, \\
G(d, s) &= \frac{d}{t^2} f(t) - \frac{s^2}{dt^2} e(t), & g_d(x) &= \int_0^x G(d, s) ds \quad (d > 0, x \in \mathbb{R}).
\end{aligned} \tag{5.14}$$

The evenness of G in s makes g_d odd. If A is the sector area and $\theta = \alpha + \beta$, substitution $s = d \tan \varphi$ in (5.13) gives

$$J - \lambda A - \mu \theta = g_d(d \tan \alpha) + g_d(d \tan \beta). \tag{5.15}$$

This identity holds for signed endpoint angles; only $|\alpha|, |\beta| < \pi/2$ and $\alpha + \beta > 0$ are required.

We choose λ, μ so that the adjusted primitive is stationary at the regular sector $(d, x) = (p, x_0)$, and set K to twice its value there. The exact choices are

$$\begin{aligned}\lambda &= \frac{1}{px_0} \left[\frac{p}{x_0} (f_0(R_0) - f_0(p)) - \frac{x_0}{p} e(R_0) \right. \\ &\quad \left. + p \int_0^{x_0} \frac{f_0(\sqrt{p^2 + s^2}) - f_0(p)}{s^2} ds + \frac{1}{p} \int_0^{x_0} e(\sqrt{p^2 + s^2}) ds \right], \\ \mu &= f_0(R_0) - \frac{1}{3} e(R_0) - \lambda R_0^2/2, \quad K = 2g_p(x_0).\end{aligned}\tag{5.16}$$

The first integrand extends continuously to $s = 0$ with value $f'_0(p)/(2p)$.

Lemma 5.2. *For the constants (5.16), $\nabla_{d,x} g_d(x) = 0$ at (p, x_0) .*

Proof. Let g_d^0 denote the primitive before subtracting area and angle. For fixed $u = \pi/6$, differentiation in the angle representation gives

$$\left. \frac{d}{dd} g_d^0(d \tan u) \right|_{d=p} = \int_0^{x_0} \frac{f'_0(t) - s^2 e'(t)/p^2}{t} ds, \quad t = \sqrt{p^2 + s^2}.$$

Set $V(s) = f_0(t) - f_0(p)$. Since $V(s) = O(s^2)$, integration by parts has no boundary term at zero and yields

$$\int_0^{x_0} \frac{f'_0(t)}{t} ds = \frac{f_0(R_0) - f_0(p)}{x_0} + \int_0^{x_0} \frac{V(s)}{s^2} ds,$$

whereas

$$\int_0^{x_0} \frac{s^2 e'(t)}{p^2 t} ds = \frac{x_0 e(R_0)}{p^2} - \frac{1}{p^2} \int_0^{x_0} e(t) ds.$$

The difference is the bracket in (5.16) divided by p , hence equals λx_0 . The subtracted area along $x = d \tan u$ is $\lambda d^2 \tan u/2$; its derivative at p is also λx_0 , and the subtracted angle is constant. Thus the derivative of $g_d(d \tan u)$ vanishes there. Finally $\partial_x g_p(x_0) = G(p, x_0) = 0$ by the definition of μ , because $x_0^2/p^2 = 1/3$. The fixed-angle derivative is $\partial_d + (x_0/p)\partial_x$, so ∂_d vanishes as well. $\square$

Lemma 5.3. *The calibrated constants satisfy*

$$W(E_\Delta) = 6K + \lambda + 2\pi\mu.\tag{5.17}$$

Proof. Take $\mathbb{T} = \mathbb{R}^2/\Lambda_\Delta$ and $N = 1$. Its regular Voronoi hexagon supplies six compatible sectors with $d = p$ and $\alpha = \beta = \pi/6$. The radial normalization and summability proved above supply the hypotheses of Proposition 4.3 with $d_0 = p$, and the torus has area one. This application uses no square-torus minimizer or incidence theorem.

On each sector $p \leq t \leq R_0$, $u_0(r) = \gamma(r)$ and $u_m(r) = (-1)^m c_m r^{k_m}$ for $r \leq t$. The edge trace and mode cancellation established above therefore give

$$H(r, \varphi) = \gamma(r) + \sum_{m \geq 1} c_m r^{k_m} \cos(k_m \varphi) = h_\Delta - C_\Delta.$$

All six normals differ by multiples of $\pi/3$, which leaves these cosines unchanged. Hence $\nabla H = \nabla h_\Delta$ and the nonnegative square in (4.11) is zero. Each sector has area $1/6$ and angle $\pi/3$; its right side in (5.15) is $2g_p(x_0) = K$. Summing the six contributions gives the cell energy $6K + \lambda + 2\pi\mu$. Lemma 2.3 identifies this cell energy with $W(E_\Delta)$, proving (5.17). $\square$

5.6 The two scalar estimates needed by the geometry

The remaining analytic task is to establish

$$\begin{aligned} K < 0, \quad 2g_d(x) \geq K \quad (d \geq b, x \geq 0), \quad b = \frac{28209}{100000}, \\ \int_0^\infty \max\{0, -G(d, s)\} ds \leq -K \quad (d \geq b). \end{aligned} \quad (5.18)$$

The rational endpoint $b = 28209/100000$ is below the geometric threshold $d_* = 1/(2\sqrt{\pi})$. Thus the domain $d \geq b$ covers all edge distances supplied by the separated-minimizer geometry; the elementary comparison is justified in Section 8.

The second estimate is essential when the perpendicular from a point to a Voronoi edge falls outside that edge. Its role is made precise in the following implication, which completes the geometric part of the argument once (5.18) has been certified.

Proposition 5.4. *Assume (5.18) for the functions and constants just constructed. Let $\mathbb{T} = \mathbb{R}^2/\Gamma$ be a flat torus of area N , where $N \geq 1$ is an integer, and let $v_1, \dots, v_N$ be distinct sites. Suppose their full periodic lift gives compatible periodic Voronoi data with lower edge distance $d_0 = b$ and at most $6N$ face-edge sectors. Let h be the mean-zero potential (4.7), with energy $\mathcal{W}_{\mathbb{T}}(h)$ defined by (4.8). Then*

$$\mathcal{W}_{\mathbb{T}}(h) \geq NW(E_\Delta).$$

Proof. The radial normalization and compact summability established above supply the radial hypotheses of Proposition 4.5 with $d_0 = b$. Evenness of G and (5.18) supply the hypotheses of Lemma 4.4 at this same lower distance. Combining its conclusion with (5.15) gives

$$J(S) \geq K + \lambda A(S) + \mu \theta(S)$$

for every sector with $d \geq b$, including the signed endpoint cases. The compatible-torus conclusion (4.15) now gives

$$\mathcal{W}_{\mathbb{T}}(h) \geq N(6K + \lambda + 2\pi\mu) = NW(E_\Delta).$$

The equality is the exact triangular calibration (5.17), not a numerical estimate of the triangular energy. $\square$

6 Infinite-mode truncation and error propagation

This section supplies analytic estimates for the modes omitted after $m = 38$. We use the radial functions b_m, U_m, u_m, B, D_m of the preceding construction, with

$$\begin{aligned} a &= \sqrt{2/\sqrt{3}}, & p &= a/2, & R_0 &= a/\sqrt{3}, \\ b &= \frac{28209}{100000}, & h_s &= \frac{3}{40}, & c &= \frac{3}{50}, & l &= R_0 + \frac{19}{250}. \end{aligned}$$

Thus $k_0 = 6$, $k_m = 6m$ for $m \geq 1$, and

$$c_m = \frac{1}{k_m} \sum_{z \in \Lambda_\Delta \setminus \{0\}} z^{-k_m}.$$

Tildes will always mean that the modes $m \geq 39$ have been omitted; the index-zero term is retained. Differences between a full quantity and its truncation are denoted by Δ . Our goal is an absolute error below 10^{-11} for f_0, e and their first two derivatives on $[b, l]$, conditional on the retained-mode bounds verified in Section 7.

6.1 Coefficient estimates

Write a lattice point as $z = a(n + q/2 + i\sqrt{3}q/2)$, with $(n, q) \in \mathbb{Z}^2$. Then

$$|z/a|^2 = n^2 + nq + q^2 \geq \frac{3}{4} \max(|n|, |q|)^2.$$

There are $8v$ integer pairs with $\max(|n|, |q|) = v$. For every $d = 6m \geq 6$ and every positive integer D , the sum outside the square $|n|, |q| \leq D$, before multiplication by a^{-d} , is therefore at most

$$8(4/3)^{d/2} \sum_{v>D} v^{1-d} \leq 8(4/3)^{d/2} \frac{D^{2-d}}{d-2}. \quad (6.1)$$

The last inequality follows by comparison with the integral from D to infinity. The first shell has six points with squared norm 1 and two with squared norm 3. Moreover,

$$\sum_{v \geq 2} v^{1-d} \leq 2^{1-d} + \int_2^\infty x^{1-d} dx = 2^{1-d} \left(1 + \frac{2}{d-2}\right) \leq 3 \cdot 2^{1-d}.$$

It follows that, for $d = k_m \geq 12$,

$$|c_m| \leq \frac{1}{da^d} \left(6 + 2 \cdot 3^{-d/2} + 24(4/3)^{d/2} 2^{1-d}\right) \leq \frac{7}{da^d}. \quad (6.2)$$

Indeed, the bracket equals $6 + 50/729 < 7$ at $d = 12$, and its two nonconstant terms decrease with d . The finite coefficient enclosures may use (6.1) with $D = 256$ for $m = 1$ and $D = 48$ for $2 \leq m \leq 38$, including the factor $1/(k_m a^{k_m})$. Reflection symmetry makes each coefficient real.

6.2 Uniform derivative bounds for omitted modes

For $k = 6m \geq 234$, define the rational number

$$w_k = 10^6 \left(30k^2(13/20)^k + 3 \cdot 10^6 k^3 2^{-k} + 5600k^5(3/5)^k + 7 \cdot 10^{11} k^2(289/500)^k\right). \quad (6.3)$$

For $0 \leq j \leq 3$ and $0 < r \leq l$, the construction satisfies

$$|u_m^{(j)}(r)| \leq w_k, \quad |D_m^{(j)}(r)| \leq w_k. \quad (6.4)$$

At spline knots these inequalities mean both one-sided derivatives. Here are explicit checks of the constants.

The polynomial cutoff $S(y) = 10y^3 - 15y^4 + 6y^5$ on $[0, 1]$, extended constantly outside that interval, obeys

$$|S| \leq 1, \quad |S'| \leq 2, \quad |S''| \leq 6, \quad |S'''| \leq 60.$$

The identities $p^4 = 1/12$ and $R_0^4 = 4/27$ give the convenient rational bounds

$$.537 < p < .538, \quad .620 < R_0 < .621, \quad .696 < l < .697, \quad l/a < 13/20, \quad R_0/a < 289/500.$$

All terminating decimals in this section denote rational numbers.

For $U_m(r) = (-1)^m c_m r^k (1 - S((r - R_0)/.1))$, Leibniz' rule and (6.2) bound the third derivative by

$$7k^2(l/a)^k \left(l^{-3} + \frac{60}{kl^2} + \frac{1800}{k^2 l} + \frac{60000}{k^3}\right).$$

Using $l > .69$ and $k \geq 234$, the coefficient of $k^2(13/20)^k$ is less than $25.445 < 30$. The same calculation for orders 0, 1, 2 gives coefficients below 0.000001, 0.000197, and 0.070572, respectively. This proves the first term inside the parentheses in (6.3).

For $p - c \leq r \leq p$, write $q_m(r) = T_{0m} + T_{1m}(r - p) + T_{2m}(r - p)^2$, so that $b_m(r) = S((r - p + c)/c)q_m(r)$. Since $p/a = 1/2$ and $p > 1/2$,

$$|T_{0m}| \leq \frac{7}{k}2^{-k}, \quad |T_{1m}| \leq 14k2^{-k}, \quad |T_{2m}| \leq \frac{14}{3}k^32^{-k}.$$

The last inequality uses $((k - 2)(k - 3)/6) - 1/2 \leq k^2/6$. Consequently

$$\begin{aligned} |q_m| &\leq (7/k + .84k + .0168k^3)2^{-k}, \\ |q'_m| &\leq (14k + .56k^3)2^{-k}, \\ |q''_m| &\leq (28/3)k^32^{-k}. \end{aligned}$$

In particular,

$$|b'''_m| \leq \frac{60}{c^3}|q_m| + \frac{18}{c^2}|q'_m| + \frac{6}{c}|q''_m| < 8500k^32^{-k} < 3 \cdot 10^6 k^3 2^{-k}.$$

The lower derivative orders satisfy the same final bound, and below $p - c$ the function is zero.

On $[p, p + .01]$, put $w = r^2 - p^2$ and $X = p + \sqrt{w}$. The polynomial formula for V_m gives, for $0 \leq \ell \leq 3$,

$$\left| \frac{d^\ell V_m}{dw^\ell} \right| \leq |c_m| k^{2\ell} X^{k-2\ell}.$$

For $j \geq \ell$, the coefficient comparison that proves this is

$$j(j-1) \cdots (j-\ell+1) \binom{k}{2j} \leq k^{2\ell} \binom{k-2\ell}{2j-2\ell}.$$

The bound $p > .537$ implies $.02p + .0001 < .04p^2$, hence $X \leq 1.2p = (3/5)a$ on this interval. Also $r < .55$ and $X > 1/2$. The chain rule gives

$$V'''_m(r) = 8r^3 \frac{d^3 V_m}{dw^3} + 12r \frac{d^2 V_m}{dw^2}.$$

Thus its absolute value is at most

$$7 \left(8(.55)^3 2^6 + \frac{12(.55)2^4}{234^2} \right) k^5 (3/5)^k < 597k^5 (3/5)^k.$$

The lower derivatives satisfy smaller bounds. The common bound $5600k^5(3/5)^k$ therefore applies through order three.

On $[p + .01, R_0]$, use $z = p + i\sqrt{r^2 - p^2}$, so that $V_m = \Re(c_m z^k)$ and $|z| = r$. Direct differentiation gives

$$|z'| \leq 7, \quad |z''| \leq 300, \quad |z'''| \leq 60000.$$

For example, $\sqrt{r^2 - p^2} > .103$, $p < .54$, and $r < .63$ suffice in the formulas $|z'| = r/\sqrt{r^2 - p^2}$, $|z''| = p^2/(r^2 - p^2)^{3/2}$, and $|z'''| = 3p^2 r/(r^2 - p^2)^{5/2}$. Since $1/r < 2$, the third derivative of $c_m z^k$ is bounded by

$$7 \left(8 \cdot 343 + \frac{12 \cdot 7 \cdot 300}{234} + \frac{2 \cdot 60000}{234^2} \right) k^2 (289/500)^k < 20000k^2 (289/500)^k.$$

All derivative orders through three are therefore bounded by $3 \cdot 10^6 k^2 (289/500)^k$ on this interval.

For the extension $\mathcal{H}V_m = V_m(R_0) + P(r - R_0)V'_m(R_0) + Q(r - R_0)V''_m(R_0)$, direct bounds on the coefficients of P, Q show, for $0 \leq j \leq 3$,

$$|P^{(j)}| \leq 2 \cdot 10^5, \quad |Q^{(j)}| \leq 4000.$$

Thus the common bound for the extension is at most

$$(1 + 200000 + 4000) 3 \cdot 10^6 k^2 (289/500)^k < 7 \cdot 10^{11} k^2 (289/500)^k.$$

It remains valid on the constant part beyond $R_0 + h_s$.

Finally the cutoff $\tau(r) = S((r - R_0 - .001)/.075)$ satisfies

$$|\tau^{(j)}| \leq 1, 27, 1067, 142223 \quad (j = 0, 1, 2, 3).$$

The maximal Leibniz factor through order three is $1 + 3 \cdot 27 + 3 \cdot 1067 + 142223 = 145506 < 10^6$. Applying this to $u_m = (1 - \tau)U_m + \tau b_m$ and $D_m = (1 - \tau)(b_m - U_m)$ proves (6.4). The series and their indicated derivatives consequently converge absolutely and locally uniformly. For $r < b$, the omitted u_m derivatives additionally have their original $O(r^{k-3})$ behavior.

6.3 Norms and bounds for retained modes

For a function F with one-sided derivatives as needed, define

$$|F|_{j,t} = \sum_{q=0}^j \frac{|F^{(q)}(t)|}{q!}, \quad \|F\|_{j,I} = \sup_{t \in I} |F|_{j,t}.$$

These norms are submultiplicative. For nonnegative weights ω_i , a weighted mode norm means $\sup_{t \in I} \sum_i \omega_i |F_i|_{j,t}$: the sum is taken before the supremum.

The finite interval verification in Section 7 establishes the following bounds for the retained modes:

$$|\tilde{B}_1|_{2,t} < 6000, \quad \sum_{i=0}^{38} (k_i + 1)^3 |\tilde{D}_i|_{3,t} < 2 \cdot 10^6, \quad b \leq t \leq l. \quad (6.5)$$

The finite coefficient enclosure also supplies $|c_1| < 1$. These assertions are not conclusions of the analytic estimates in this section. Conditional on them, the remaining estimates below are independent of the finite calculation.

On $[0, b]$,

$$\tilde{B}_1(t) = \pi t + \sum_{m=1}^{38} (-1)^m k_m c_m t^{k_m-1}$$

has norm $|\tilde{B}_1|_{2,t} < 10$. For an explicit check, use $t < b < 2/7$, $a > 1$, $\pi < 22/7$, and $|c_1| < 1$ for the first mode. The modes $k \geq 12$ contribute at most $\sum_{k=12,18,\dots} 7k^2 (2/7)^{k-3} < .014$; the first term and its geometrically decreasing ratios prove this last rational inequality. Together with the linear term and the $k = 6$ term this is less than 10.

6.4 Summing the omitted modes

Set

$$\mathfrak{a} = \sum_{k=234,240,\dots} (k+1)^4 w_k < 2 \cdot 10^{-21}, \quad W_* = 4 \sum_{k=234,240,\dots} w_k \leq 4\mathfrak{a}. \quad (6.6)$$

Here is a rational verification of the numerical inequality. For each term $(k+1)^4 k^s r^k$, with

$$(s, r) = (2, 13/20), (3, 1/2), (5, 3/5), (2, 289/500),$$

the ratio at the next allowed index is

$$\left(\frac{k+7}{k+1}\right)^4 \left(\frac{k+6}{k}\right)^s r^6.$$

It decreases with k , and at $k = 234$ it is strictly below $1/10$ for all four pairs. The first weighted summand obeys the exact rational comparison

$$235^4 10^6 (30 \cdot 234^2 (13/20)^{234} + 3 \cdot 10^6 \cdot 234^3 2^{-234} + 5600 \cdot 234^5 (3/5)^{234} + 7 \cdot 10^{11} \cdot 234^2 (289/500)^{234}) < 1.56 \cdot 10^{-21}.$$

Multiplying by $10/9$ gives a bound less than $2 \cdot 10^{-21}$, as asserted in (6.6).

Since $\Delta B_1 = \sum_{m \geq 39} u'_m$, the derivative estimates give

$$|\Delta B_1|_{2,t} \leq W_*, \quad 0 \leq t \leq l,$$

and, on $[b, l]$,

$$\sum_{i \geq 39} (k_i + 1)^j |D_i|_{3,t} \leq 4a, \quad 0 \leq j \leq 4.$$

Indeed the divided-factorial sums use factors $5/2$ and $8/3$, respectively, both less than 4.

6.5 Propagation to the sector functions

Recall the quantities entering the sector formula:

$$\begin{aligned} B_1(t) &= B'(t) + 1/t, & S_1(t) &= \int_0^t B(r) r \, dr, \\ S_2(t) &= \int_0^t (-2B_1(r) + rB_1(r)^2) \, dr, & H_k(t) &= t^{-k} \int_0^t B_1(r) r^k \, dr, \\ Q_i(t) &= tD'_i(t) - k_i D_i(t). \end{aligned}$$

Write $N = 2 \cdot 10^6$ temporarily. Direct differentiation of ΔS_1 gives

$$|\Delta S_1|_{2,t} \leq 3W_*.$$

For example $|\Delta B^{(j)}| \leq W_*/4$ through order three, so the norm is at most $[t^2/2 + t + (1+t)/2]W_*/4$ for $t \leq l < 1$. For the other integral, use

$$\Delta(-2B_1 + tB_1^2) = (-2 + t(2\tilde{B}_1 + \Delta B_1))\Delta B_1.$$

The second-order norm of its integral is at most twice the supremum of the first-order norm of the integrand, because $l < 1$. Submultiplicativity and $|t|_{1,t} < 2$ yield

$$|\Delta S_2|_{2,t} \leq (4 + 8 \cdot 6000 + 4W_*)W_* < 60000W_*.$$

The identities and elementary bounds

$$H'_k = B_1 - kH_k/t, \quad |H_k| \leq \frac{t\|B_1\|_{0,[0,t]}}{k+1}, \quad |H'_k| \leq 2\|B_1\|_{0,[0,t]}, \quad |H''_k| \leq |B'_1| + \frac{2k}{t}\|B_1\|_{0,[0,t]}$$

imply, for $b \leq t \leq l$,

$$|H_k|_{2,t} \leq 6(k+1)(6000 + W_*), \quad |\Delta H_k|_{2,t} \leq 6(k+1)W_*. \quad (6.7)$$

Indeed $l/(k+1) + 5/2 + k/b < 6(k+1)$. The bound for ΔH_k follows by the same linear argument applied to ΔB_1 .

The radial source and energy functions are

$$\begin{aligned} f_0(t) &= -2\pi \left(S_1(t) + t^2 \sum_i \frac{D_i(t)}{k_i + 2} \right) \\ &\quad - \frac{1}{2} \left(\log t + S_2(t) + \sum_i (-2D_i + 2k_i D_i H_{k_i})(t) + \sum_{i,j} \frac{k_i k_j}{k_i + k_j} D_i(t) D_j(t) \right), \\ e(t) &= \frac{1}{2} \sum_{i,j} \frac{Q_i(t) Q_j(t)}{k_i + k_j}. \end{aligned}$$

The preceding bounds give

$$\begin{aligned} |f_0 - \tilde{f}_0|_{2,t} &\leq 2\pi(3W_* + 16\mathfrak{a}) \\ &\quad + \frac{1}{2}(60000W_* + 8\mathfrak{a} + 12(2 \cdot 10^6)W_* + 48(6000 + W_*)\mathfrak{a} \\ &\quad + 8(2 \cdot 10^6)\mathfrak{a} + 16\mathfrak{a}^2) < 6 \cdot 10^7 \mathfrak{a}. \end{aligned} \quad (6.8)$$

For completeness, $|t^2|_{2,t} = (t+1)^2 < 4$ gives the $16\mathfrak{a}$ source term. The sum of $-2D_i$ contributes $8\mathfrak{a}$. Split the mixed term as

$$\sum_i k_i (D_i H_{k_i} - \tilde{D}_i \tilde{H}_{k_i}) = \sum_{i \leq 38} k_i \tilde{D}_i \Delta H_{k_i} + \sum_{i \geq 39} k_i D_i H_{k_i}.$$

The finite and omitted weighted norms give $12NW_*$ and $48(6000 + W_*)\mathfrak{a}$ after including its factor 2. Bounding $k_i k_j / (k_i + k_j)$ by $k_i k_j$, the quadratic difference is at most $8N\mathfrak{a} + 16\mathfrak{a}^2$. These are precisely the terms in (6.8). Finally $W_* \leq 4\mathfrak{a}$, $\pi < 22/7$, and $\mathfrak{a} < 2 \cdot 10^{-21}$ make its total coefficient less than $56,264,181 < 6 \cdot 10^7$.

Since $k_i \geq 6$ and $t < 1$, direct differentiation of Q_i shows

$$|Q_i|_{2,t} \leq 2(k_i + 1)|D_i|_{3,t}.$$

For example, its coefficients on $|D_i|$, $|D'_i|$, $|D''_i|$, $|D'''_i|$ are bounded respectively by k_i , $t + k_i - 1$, $t + (k_i - 2)/2$, and $t/2$. The finite and omitted sums of these norms are bounded by $2N$ and $8\mathfrak{a}$. Dropping the denominator $k_i + k_j \geq 1$ in the energy difference therefore gives

$$|e - \tilde{e}|_{2,t} \leq 16(2 \cdot 10^6)\mathfrak{a} + 32\mathfrak{a}^2. \quad (6.9)$$

The norm contains half the absolute second derivative. Accounting for this factor, (6.8) and (6.9) imply absolute truncation errors less than 10^{-11} for every member of

$$f_0, \quad e, \quad f'_0, \quad e', \quad f''_0, \quad e'' \quad (b \leq t \leq l). \quad (6.10)$$

In fact the respective second derivative bounds are below $2.4 \cdot 10^{-13}$ and $1.3 \cdot 10^{-13}$.

6.6 The constant region

For $r \geq l$, all D_i vanish and B is constant. The omitted constant satisfies

$$|\Delta B(l)| \leq \sum_{k=234,240,\dots} \frac{7}{k} (R_0/a)^k \leq \frac{(7/234)(289/500)^{234}}{1 - (289/500)^6} < 6.075 \cdot 10^{-58} < 10^{-11}.$$

Since $B_1(t) = 1/t$ there, the derivatives of $\log t$ and $S_2(t)$ cancel. The full infinite-mode functions consequently obey the exact formula

$$e(t) = 0, \quad f_0(t) = f_0(l) - \pi B(l)(t^2 - l^2), \quad t \geq l. \quad (6.11)$$

Thus outside $[b, l]$ one propagates the enclosures of $f_0(l)$ and $B(l)$ through this exact formula; no uniform absolute error for arbitrarily large t is asserted.

7 The finite scalar verification

The geometric argument requires the two inequalities (5.18) for the exact functions G and g_d . We now specify a finite calculation that implies them. Its inputs are the radial data (5.7), the formulas (5.11), and the constants defined by the integrals (5.16). Its outputs are interval bounds for these constants, two strict bounds on a compact rectangle, and a positive Hessian bound near the exact stationary point. The estimates of Section 6 connect the finite radial calculation to the infinite series.

Every terminating decimal in this section denotes an exact rational number. An interval enclosure always contains the exact mathematical quantity in question. In particular, the constants λ, μ, K are not replaced by rounded numerical choices. Their enclosures are used in all subsequent operations, while Lemma 5.2 supplies stationarity for their exact values.

7.1 Outward arithmetic and derivative intervals

The calculation follows the interval inclusion principle: every operation encloses its exact result, and analytic remainders are added explicitly. Revol and Rouillier [15] discuss this principle and the dependence of enclosure width on expressions and subdivision. We specify the integer operations here, so the proof's arithmetic assumptions and rounding rules are explicit.

Put $q = 2^{144}$. A scalar interval is represented by integers $u \leq v$, meaning $[u/q, v/q]$. Rational inputs are rounded down at the lower endpoint and up at the upper endpoint. Addition and negation are exact at this denominator. Multiplication takes the least and greatest of the four endpoint products, then divides by q with outward rounding. If $0 < u \leq v$, inversion gives

$$\left[\frac{\lfloor q^2/v \rfloor}{q}, \frac{\lceil q^2/u \rceil}{q} \right];$$

negative intervals are treated by negation, and division by an interval containing zero is forbidden. Integer powers use repeated multiplication. For a positive interval, integer square roots give the enclosure

$$\left[\frac{\lfloor \sqrt{uq} \rfloor}{q}, \frac{\lfloor \sqrt{vq} \rfloor + 1}{q} \right].$$

The extra unit in the upper endpoint is harmless when the upper square root is an integer.

For logarithms, first extract an integer power of two and reduce the argument to $[1/4, 1]$. The arguments used below are positive table intervals or enclosures of fixed positive constants; their widths allow this reduction. With $z = (v - 1)/(v + 1)$, evaluate

$$\log v = 2 \sum_{j=0}^{199} \frac{z^{2j+1}}{2j+1} + \mathcal{R}, \quad |\mathcal{R}| \leq \frac{2(3/5)^{401}}{401(1 - (3/5)^2)} < q^{-1}. \quad (7.1)$$

The same series at $z = 1/3$ encloses $\log 2$, used to restore the extracted power. All summation errors are already enclosed by outward arithmetic; the displayed remainder is added separately. Similarly, use

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239), \quad \arctan(1/v) = \sum_{j=0}^{149} \frac{(-1)^j v^{-2j-1}}{2j+1} + \mathcal{R}_v. \quad (7.2)$$

The alternating-series bound gives $|\mathcal{R}_v| \leq v^{-301}/301 < q^{-1}$ for both values of v . Adding $[-2q^{-1}, 2q^{-1}]$ to each arctangent enclosure before the linear combination is sufficient. These procedures also enclose p, R_0, x_0, l using their exact defining formulas.

To bound derivatives we use interval jets

$$(F_0, F_1, F_2, F_3), \quad F_j \supseteq F^{(j)}(t)/j!.$$

Addition, multiplication, inversion, logarithms, and square roots are performed as formal power series through degree three. For example, the product coefficient is $\sum_{r=0}^j F_r G_{j-r}$, and the inverse coefficients follow recursively from the product being one. A variable on a radial interval T is represented by $(T, 1, 0, 0)$; the resulting coefficients enclose its derivatives at every point of T on the chosen spline piece. At a single point, the leading interval encloses that point and the same derivative entries are used. After differentiation only the coefficients supported by the available derivative orders are used. The calculation below never requires a fourth derivative of a spline: third derivatives of B, D_i suffice for the second derivatives of f_0, e and for the radial quadrature errors.

For a differential equation $Y' = F(t, Y)$ with an enclosed value at the evaluation point, its jet coefficients are reconstructed in increasing order by

$$Y_{j+1} = \frac{[F(t, Y)]_j}{j+1}, \quad 0 \leq j \leq 2. \quad (7.3)$$

Thus the enclosed value and the known differential equation supply the derivatives without numerical differentiation.

Spline polynomials are evaluated by interval Horner arithmetic. One may first translate a polynomial $P(t) = \sum_{\ell=0}^N a_\ell t^\ell$ to the midpoint m of its leading interval:

$$P(m+h) = \sum_{j=0}^N \left(\sum_{\ell=j}^N \binom{\ell}{j} a_\ell m^{\ell-j} \right) h^j. \quad (7.4)$$

This is an exact polynomial identity, also for jets. It reduces interval widths without changing the spline or introducing an approximation; direct Horner evaluation remains a valid enclosure.

7.2 Coefficients and the radial table

Retain the auxiliary mode 0 and modes $1 \leq m \leq 38$. The two modes with exponent six remain separate in sums involving D_i . Enclose each c_m by summing over $|n|, |r| \leq D$, where $D = 256$ for $m = 1$ and $D = 48$ otherwise, and then adding the symmetric error in (6.1),

multiplied by a^{-k_m}/k_m . The summands can be evaluated using only integers before that last factor: if $k = k_m$, $A = 2n + r$, and $N = n^2 + nr + r^2$, then

$$\operatorname{Re}(n + r/2 + i\sqrt{3}r/2)^{-k} = \frac{\sum_{j=0}^{k/2} (-1)^j \binom{k}{2j} A^{k-2j} (3r^2)^j}{2^k N^k}. \quad (7.5)$$

The pair $(0, 0)$ is omitted. The resulting enclosure also tests $|c_1| < 1$.

The radial breakpoints are

$$b, \quad p - .06, \quad p, \quad R_0, \quad R_0 + .001, \quad R_0 + .075, \quad l = R_0 + .076. \quad (7.6)$$

Each intervening interval is subdivided by affine interpolation between its exact endpoints. For the first interval use $n = \lfloor 8000(u - v) \rfloor + 1$ subintervals, and for all later intervals use $n = \lfloor 24000(u - v) \rfloor + 1$, where v, u are its endpoints. In execution an outward upper bound on the product is used to choose n , which can only add subdivisions. All true node positions are enclosed by the corresponding interval arithmetic. The length of every actual subinterval is bounded above in the calculation; these upper bounds are used in all error estimates.

For the fixed table, write $[u_i/q, v_i/q]$ for the valid outward enclosure of its i th node, $0 \leq i \leq m$. The interval constructor enforces $u_i \leq v_i$, and the node arithmetic preserves this invariant. After appending the last node and before propagating any integral state, the program requires

$$v_i < u_{i+1} \quad (0 \leq i < m); \quad (7.7)$$

it throws an exception if a comparison fails. Thus, on a completed table construction, $u_i \leq v_i < u_{i+1} \leq v_{i+1}$. Both endpoint arrays are strictly increasing, and the stored step interval is

$$H_i = [(u_{i+1} - v_i)/q, (v_{i+1} - u_i)/q]$$

with positive lower endpoint. This also justifies every division by H_i below. These facts concern the initialized table and its unchanged aligned node, point, and cell arrays in the fixed calculation.

On $0 < t \leq p - .06$, the truncated functions have the monomial form

$$\tilde{B}(t) = -\log t + \pi t^2/2 + \sum_{m=1}^{38} a_m t^{k_m}, \quad \tilde{B}_1(t) = \pi t + \sum_{m=1}^{38} k_m a_m t^{k_m-1}, \quad a_m = (-1)^m c_m.$$

For compact formulas introduce the finite list

$$(e_0, v_0) = (2, \pi), \quad (e_m, v_m) = (k_m, k_m a_m) \quad (1 \leq m \leq 38).$$

This list indexes monomials of $\tilde{B}_1$, rather than the sector modes; its exponent $e_0 = 2$ is unrelated to $k_0 = 6$. The starting integrals, including their derivatives, are obtained from the exact identities

$$\begin{aligned} \tilde{S}_1(t) &= \frac{t^2}{2} \left(\frac{1}{2} - \log t \right) + \frac{\pi}{8} t^4 + \sum_{m=1}^{38} \frac{a_m t^{k_m+2}}{k_m + 2}, \\ \tilde{S}_2(t) &= -2 \sum_{i=0}^{38} \frac{v_i t^{e_i}}{e_i} + \sum_{i,j=0}^{38} \frac{v_i v_j t^{e_i+e_j}}{e_i + e_j}, \\ \tilde{H}_k(t) &= \sum_{i=0}^{38} \frac{v_i t^{e_i}}{e_i + k}. \end{aligned} \quad (7.8)$$

In particular, the table starts propagation at the exact node $p - .06$ with these values; preceding nodes and cells use (7.8) directly.

There is a useful exact simplification for the sector functions on this first interval. Since $b_m(t) = 0$, $D_m(t) = -a_m t^{k_m}$ and $D_0(t) = 0$, substitution in (5.9) cancels every nonconstant mode and gives $H(r, \varphi) = \gamma(r)$ for $r \leq t$. Equivalently, every Q_i vanishes. Direct radial integration gives

$$f_0(t) = (\pi t^2 - \tfrac{1}{2}) \log t - \frac{3\pi^2 t^4}{8}, \quad e(t) = 0, \quad 0 < t \leq p - .06. \quad (7.9)$$

Indeed,

$$\int_0^t \gamma(r) r \, dr = -\frac{t^2}{2} \log t + \frac{t^2}{4} + \frac{\pi t^4}{8}, \quad \int_0^t \text{ren} \gamma'(r)^2 r \, dr = \log t - \pi t^2 + \frac{\pi^2 t^4}{4}.$$

Combining these with the source and energy factors in the sector formula proves (7.9). The identity holds for the full infinite series as well as for every finite truncation. It may therefore replace the mode sums when evaluating f_0, e and their first two derivatives on the first interval, producing tighter enclosures. The full finite-mode values (7.8) at $p - .06$ are still required to start the continuation of S_1, S_2, H_k .

For a subsequent radial step $[v, u]$, propagate the integrals by

$$\begin{aligned} \tilde{S}_1(u) &= \tilde{S}_1(v) + \int_v^u r \tilde{B}(r) \, dr, \\ \tilde{S}_2(u) &= \tilde{S}_2(v) + \int_v^u (-2\tilde{B}_1(r) + r \tilde{B}_1(r)^2) \, dr, \\ \tilde{H}_k(u) &= (v/u)^k \tilde{H}_k(v) + \int_v^u (r/u)^k \tilde{B}_1(r) \, dr. \end{aligned} \quad (7.10)$$

For each integrand Y , the midpoint rule has the rigorous enclosure

$$\int_v^u Y(r) \, dr \in (u-v)Y((v+u)/2) + \left[-\frac{(u-v)^3}{24} \sup_{[v,u]} |Y''|, \frac{(u-v)^3}{24} \sup_{[v,u]} |Y''| \right]. \quad (7.11)$$

Midpoint values use point jets; second-derivative bounds use a jet whose leading interval contains $[v, u]$. The formula applies to weak second derivatives bounded almost everywhere as well.

For values inside a step, use the integral equations rather than the midpoint approximation. If $T = [v, u]$ and $h = u - v$, then

$$\begin{aligned} \tilde{S}_j(T) &\subseteq \tilde{S}_j(v) + [0, h] \text{ range}_T Y_j, \quad j = 1, 2, \\ \tilde{H}_k(T) &\subseteq (v/T)^k \tilde{H}_k(v) + [-h \|\tilde{B}_1\|_{0,T}, h \|\tilde{B}_1\|_{0,T}], \end{aligned} \quad (7.12)$$

where $Y_1 = r \tilde{B}$ and $Y_2 = -2\tilde{B}_1 + r \tilde{B}_1^2$. For the last inclusion, $(r/t)^k \leq 1$ whenever $v \leq r \leq t \leq u$. Recover the positive jet orders from

$$\tilde{S}'_1 = t \tilde{B}, \quad \tilde{S}'_2 = -2\tilde{B}_1 + t \tilde{B}_1^2, \quad \tilde{H}'_k = \tilde{B}_1 - k \tilde{H}_k/t. \quad (7.13)$$

Substitution into (5.11) produces both node enclosures and cell enclosures of the ordered six-tuple

$$(\tilde{f}_0, \tilde{f}'_0, \tilde{f}''_0, \tilde{e}, \tilde{e}', \tilde{e}''). \quad (7.14)$$

No numerical differentiation of table values is used.

Separately, subdivide each interval in (7.6) into steps of length at most .00022. Direct interval jets for $\tilde{B}_1$ and $\tilde{D}_i$ test the two finite-core inequalities (6.5). Together with $|c_1| < 1$,

these are the finite premises of the truncation analysis; their verification does not use any infinite-mode error bound. Only after these tests pass is (6.10) invoked.

The spline data are C^2 and have bounded one-sided third derivatives. Accordingly f_0, e are C^1 with locally bounded weak second derivatives on (b, ∞) . A radial cell uses its own spline formula at its endpoints; either one-sided second derivative is valid for its interior. When a query meets a knot, the hull includes both adjacent cells. Thus no jump of a second derivative is mistaken for global classical differentiability.

7.3 Queries for the infinite-mode functions

Let $\epsilon = [-10^{-11}, 10^{-11}]$, and let T be a valid outward enclosure of the queried true radii. In the fixed production calls every such radius is at least b , although T may extend below b . For derivatives, take the componentwise hull of (7.14) over every radial cell meeting T and add ϵ to each component. By (6.10), this encloses the full infinite-mode tuple $(f_0, f'_0, f''_0, e, e', e'')$ at every queried true radius in $T \cap [b, l]$. The exact continuation below covers queried true radii at least l . Here is the cell-selection argument for the implementation. In integer endpoint units write the valid query as $[a, b']$, let j be the first index with $v_j \geq a$, and let k be the first index with $u_k > b'$, using $m + 1$ if either index is absent. The closed envelope of cell i is $[u_i, v_{i+1}]$. It meets $[a, b']$ exactly when $v_{i+1} \geq a$ and $u_i \leq b'$. By the ordering proved above, these conditions are equivalent to $i \geq j - 1$ and $i < k$. Therefore the binary searches on the upper and lower endpoint arrays select exactly

$$\text{im} = \max(0, j - 1) \leq i < \min(m, k) = \text{ix}.$$

The non-strict intersection inequalities include both cells at a shared endpoint. Every true cell meeting the query lies in its envelope and is therefore selected; any extra cells only widen the hull. This is a statement about the fixed initialized table, not a general-purpose lookup interface.

For a query requiring values alone, improve the enclosure by linear interpolation of the endpoint values. On a radial cell $[v, u]$ put

$$X = [0, 1] \cap \frac{T - v}{u - v}, \quad L_Y(T) = Y(v) + X(Y(u) - Y(v)), \quad Y = \tilde{f}_0, \tilde{e}.$$

An empty intersection is discarded. The interpolation error is

$$|Y(t) - L_Y(t)| \leq \frac{(u - v)^2}{8} \sup_{[v, u]} |Y''|. \quad (7.15)$$

The cell enclosure supplies the derivative bound. Add this symmetric error and ϵ , then take the hull over all relevant cells. Shared coefficient uncertainties need not be independent: interval operations enclose all their possible values and hence also their correlated exact values. In particular, the positive stored interval H_i contains the true step $u - v$, so outward evaluation of the quotient defining X contains the normalized coordinate of every true queried point in that cell. Intersecting with $[0, 1]$ cannot discard such a point.

Queries that meet $[l, \infty)$ also include the exact continuation (6.11). Enclose its two constants by the last node value plus ϵ and by

$$B(l) \in -\log R_0 + \pi R_0^2/2 + \sum_{m=1}^{38} V_m(R_0) + \epsilon. \quad (7.16)$$

Thus the continuation contributes the tuple

$$(f_0(l) - \pi B(l)(T^2 - l^2), -2\pi B(l)T, -2\pi B(l), 0, 0, 0). \quad (7.17)$$

Using the entire query interval in this expression is conservative even if it extends below l , since the relevant exterior subset is contained in it. This rule propagates the constant uncertainties through the exact formula and asserts no radius-independent error on the unbounded tail.

7.4 Enclosing the calibrated constants

We evaluate the exact integrals defining λ in (5.16). Put $s_0 = .01$ and $t(s) = \sqrt{p^2 + s^2}$. Taylor's formula at p removes the apparent division by zero in the first integral. Specifically,

$$\begin{aligned} \int_0^{s_0} p \frac{f_0(t(s)) - f_0(p)}{s^2} ds \in f'_0(p) \left(\frac{s_0}{2} - \frac{s_0^3}{24p^2} + \left[0, \frac{s_0^5}{80p^4} \right] \right) \\ + \frac{s_0^3}{6} \left[\frac{p}{(\sqrt{p^2 + s_0^2} + p)^2}, \frac{1}{4p} \right] \text{range}_{[p, t(s_0)]} f''_0. \end{aligned} \quad (7.18)$$

Indeed, $t - p = s^2/(t + p)$, while

$$\frac{p}{t + p} = \frac{1}{2} - \frac{s^2}{8p^2} + \left[0, \frac{s^4}{16p^4} \right].$$

The quadratic Taylor remainder of f_0 supplies the second line of (7.18). The derivative range comes from the radial cells with the infinite-mode error added.

The implementation obtains this range by a direct cell scan over $[p, t(s_0)]$, and obtains the derivative bounds below by a second direct scan over $[p, R_0]$. Both use the same closed-cell selection rule, with outward endpoint bounds, but neither adds the exterior formula. This is sufficient here: $p^4 = 1/12$ gives $p > 1/2 > b$, $x_0 = R_0/2 > s_0$, and

$$t(x_0) = \sqrt{p^2 + (R_0/2)^2} = R_0 < l.$$

Hence both true scan domains are contained in the finite table $[b, l]$. A nonempty scan alone would not justify a query extending beyond that table.

On $[s_0, x_0]$ integrate $p(f_0(t(s)) - f_0(p))/s^2$ by a midpoint sum with 45000 equal steps. Integrate $e(t(s))/p$ on $[0, x_0]$ by a separate midpoint sum with 45000 equal steps. Obtain integer upper bounds

$$C \geq \sup_{[p, R_0]} |f'_0|, \quad C' \geq \sup_{[p, R_0]} |f''_0|, \quad E \geq \sup_{[p, R_0]} |e'|, \quad E' \geq \sup_{[p, R_0]} |e''|$$

from the cell enclosures plus ϵ . The two integrands have second derivatives bounded respectively by

$$\frac{C'}{p} + \frac{8C}{s_0^2}, \quad \frac{E'}{p} + \frac{E}{p^2}. \quad (7.19)$$

For the first bound, set $V = f_0(t(s)) - f_0(p)$; then

$$|V| \leq Cs^2/(2p), \quad V'' = f''_0(t)(s/t)^2 + f'_0(t)p^2/t^3,$$

and differentiate pV/s^2 twice. The resulting terms are $p(V''/s^2 - 4V'/s^3 + 6V/s^4)$, bounded by the first expression in (7.19). The second follows by twice differentiating $e(t(s))/p$. For an interval of length L and equal step length h , add the symmetric error $Lh^2 \sup |Y''|/24$ to each complete midpoint sum. Together with (7.18), the endpoint terms and the factors in (5.16), this encloses λ . The same formula then encloses μ .

To enclose $K = 2 \int_0^{x_0} G(p, s) ds$, choose an integer n and equal steps of length $h = x_0/n \leq .0001$, and use their midpoint sum. Independently enclose $G_{ss}(p, s)$ on a partition with step length at most .0002. If h is the midpoint step length and L_K bounds this second derivative, the enclosure is

$$K \in 2 \left(\sum_j h G(p, s_j) + [-x_0 h^2 L_K / 24, x_0 h^2 L_K / 24] \right). \quad (7.20)$$

The factor two multiplies both the sum and its error.

The first sign tests are

$$\begin{aligned} -6.731 < \lambda < -6.729, \quad K < 0, \\ f_0(l) - \lambda l^2 / 2 - \mu > 0, \quad -\pi B(l) - \lambda / 2 > 0. \end{aligned} \quad (7.21)$$

Each strict inequality is accepted only when the corresponding interval endpoint proves it. The bound on λ also serves as a check on the normalization. The last two signs and (6.11) give

$$f(t) = f(l) + (-\pi B(l) - \lambda / 2)(t^2 - l^2) > 0, \quad G(d, s) > 0 \quad \text{if } t = \sqrt{d^2 + s^2} \geq l. \quad (7.22)$$

Here $e(t) = 0$ and $d/t^2 > 0$.

7.5 The compact rectangle and its derivative bounds

Because $l < .697$ and $\sqrt{b^2 + .65^2} > l$, the signs (7.22) reduce the remaining work to

$$\mathcal{D} = [b, .697] \times [0, .65]. \quad (7.23)$$

For $d \geq .697$, $G(d, s) > 0$ for every $s \geq 0$ and $g_d(x) \geq 0$. For $d \geq b$ and $x \geq .65$, the primitive is increasing in x . The negative part of G vanishes for $s \geq .65$.

We obtain derivatives of G by differentiating its defining formula, using the full radial query enclosures. The following identities specify all needed factors. Put

$$t = \sqrt{d^2 + s^2}, \quad A = d/t^2, \quad Z = -s^2/(dt^2) = A - 1/d, \quad r_d = d/t, \quad r_s = s/t,$$

and write $f' = f'_0 - \lambda t$, $f'' = f''_0 - \lambda$. Then

$$\begin{aligned} A_d &= (1 - 2d^2/t^2)/t^2, & Z_d &= A_d + 1/d^2, \\ A_s &= -2ds/t^4, & Z_s &= A_s, \\ A_{dd} &= 2d(d^2 - 3s^2)/t^6, & Z_{dd} &= A_{dd} - 2/d^3, \\ A_{ss} &= -2d/t^4 + 8ds^2/t^6, & Z_{ss} &= A_{ss}. \end{aligned} \quad (7.24)$$

For $v = d$ or $v = s$ let r_v have the meaning above. The derivative formulas are

$$\begin{aligned} G_v &= A_v f + A f' r_v + Z_v e + Z e' r_v, \\ G_{vv} &= A_{vv} f + 2A_v f' r_v + A(f'' r_v^2 + f'(1 - r_v^2)/t) \\ &\quad + Z_{vv} e + 2Z_v e' r_v + Z(e'' r_v^2 + e'(1 - r_v^2)/t). \end{aligned} \quad (7.25)$$

These formulas hold on each smooth piece and almost everywhere across the radial knots. No mixed second derivative of G is needed.

Partition both sides of (7.23) into intervals of length at most .0008. Interval substitution in (7.25) on every closed rectangle supplies global upper bounds

$$L_1 \geq \|G_d\|_{\infty, \mathcal{D}}, \quad L_2 \geq \|G_s\|_{\infty, \mathcal{D}}, \quad L_3 \geq \|G_{dd}\|_{\infty, \mathcal{D}}, \quad L_4 \geq \|G_{ss}\|_{\infty, \mathcal{D}}. \quad (7.26)$$

The value-only interpolation is not used to obtain these derivative bounds.

7.6 The primitive away from its stationary point

Let

$$\mathcal{D}_0 = [.52528, .54930] \times [.29420, .32621]. \quad (7.27)$$

Its interior contains (p, x_0) . To verify the primitive bound on the complement, partition the d interval at the two vertical sides of $\mathcal{D}_0$, and partition the s interval at its two horizontal sides. Subdivide these pieces into steps with maximal lengths

$$D_h \leq .00019, \quad S_h \leq .00018.$$

Every cell therefore lies either in $\mathcal{D}_0$ or outside its interior. For each d node, form successive midpoint sums from 0 to every x node, using the actual step lengths:

$$Q(d, x) = \sum_{[s_j, s_{j+1}] \subseteq [0, x]} (s_{j+1} - s_j) G\left(d, \frac{s_j + s_{j+1}}{2}\right).$$

The sum at $x = 0$ is zero and is tested as well. The value queries in Section 7.3 may be used here. At every node outside the interior of $\mathcal{D}_0$, enclose $2Q(d, x) - K$ and let m_* be a lower bound for all these enclosures.

The quadrature error at grid nodes is bounded by E_0 , and the additional interpolation error between nodes by E_I , where

$$E_0 = \frac{2(.65)S_h^2 L_4}{24}, \quad E_I = \frac{2(D_h^2(.65)L_3 + S_h^2 L_2)}{8}, \quad (7.28)$$

Indeed, the bilinear interpolation estimate follows from

$$\partial_d^2(2g_d(x) - K) = 2 \int_0^x G_{dd}(d, s) ds, \quad \partial_x^2(2g_d(x) - K) = 2G_s(d, x).$$

Apply the one-dimensional interpolation error in each variable successively; the resulting bounds add and require no mixed derivative. Thus the finite assertion

$$m_* - E_0 - E_I > 0 \quad (7.29)$$

proves $2g_d(x) > K$ throughout $\mathcal{D} \setminus \text{int } \mathcal{D}_0$. All the derivative estimates are weak derivative estimates, so the same conclusion holds when a quadrature segment or rectangle crosses a radial knot.

7.7 The negative part on the whole rectangle

At every d node, including those between the vertical sides of $\mathcal{D}_0$, use the same s partition to enclose

$$N(d) = \sum_j (s_{j+1} - s_j) \max \left\{ 0, -G\left(d, \frac{s_j + s_{j+1}}{2}\right) \right\}.$$

Let N_* be an upper bound for all these sums. The function $y \mapsto \max\{0, -y\}$ is 1-Lipschitz. On a subinterval of length h , its midpoint error is therefore at most $L_2 h^2/4$: integrate $L_2|s - s_{\text{mid}}|$. Summing the errors gives at most $.65S_h L_2/4$. Every d lies within $D_h/2$ of a node; changing d to that node changes the integral by at most $.65D_h L_1/2$. Consequently the assertion

$$N_* + K + .65 \left(\frac{S_h L_2}{4} + \frac{D_h L_1}{2} \right) < 0 \quad (7.30)$$

proves

$$\int_0^{.65} \max\{0, -G(d, s)\} ds < -K \quad (b \leq d \leq .697).$$

Together with (7.22), this gives the entire negative-part inequality in (5.18).

7.8 Convexity in the stationary rectangle

It remains to prove the primitive inequality on $\mathcal{D}_0$. Partition its d interval into steps of length at most .00004. Partition $[0, .32621]$ into steps of length at most .00008, including .29420 as a node. The accumulation in s must begin at zero even though the Hessian is tested only above .29420.

For a d interval D and an s interval S_j , use the full derivative queries to enclose

$$F_1(D, S_j) \supseteq G_{dd}(D, S_j), \quad F_2(D, S_j) \supseteq G_d(D, S_j), \quad F_3(D, S_j) \supseteq G_s(D, S_j).$$

For a cell $D \times S_j$ meeting $\mathcal{D}_0$, enclosures of the three Hessian entries of $(d, x) \mapsto g_d(x)$ are

$$\begin{aligned} A_j &= \sum_{i < j} |S_i| F_1(D, S_i) + [0, |S_j|] F_1(D, S_j), \\ B_j &= F_2(D, S_j), \quad C_j = F_3(D, S_j). \end{aligned} \tag{7.31}$$

The first line is direct interval integration, not midpoint quadrature: it bounds both the completed subintervals and the partial final subinterval. Test, on every such cell, that

$$\inf A_j > 0, \quad \inf C_j > 0, \quad \inf(A_j C_j - B_j^2) > 0. \tag{7.32}$$

Interval multiplication encloses the determinant even when its factors are correlated. These inequalities imply positive definiteness, and in particular positive semidefiniteness, of the Hessian almost everywhere on $\mathcal{D}_0$.

The function $(d, x) \mapsto g_d(x)$ belongs to $W^{2,\infty}$ on a neighbourhood of this rectangle. To justify the passage from Hessian bounds to convexity, mollify on each smaller interior rectangle: the mollified Hessian is positive semidefinite, so the mollified functions are convex. Local uniform convergence proves convexity in the interior, and continuity includes the boundary. By Lemma 5.2, (p, x_0) is an interior stationary point. Hence

$$2g_d(x) \geq 2g_p(x_0) = K \quad ((d, x) \in \mathcal{D}_0). \tag{7.33}$$

This step uses the exact stationary point and does not infer convexity or stationarity from sampled function values.

7.9 Finite assertions and the scalar conclusion

For clarity, the finite procedure has the following fixed parameters and obligations.

Quantity	Parameter or required assertion
Arithmetic	Denominator 2^{144} , jet order 3
Retained modes	$i = 0, 1, \dots, 38$
Coefficient squares	$D = 256$ for $m = 1$, $D = 48$ otherwise
Core mesh	At most .00022 on each spline interval
Core assertions	$ c_1 < 1$ and (6.5)
Radial table	Densities 8000 and 24000 as specified above
Radial node ordering	Strict separation (7.7) at every adjacent pair
λ integrals	45000 midpoint steps each; $s_0 = .01$
K integral	Steps at most .0001; derivative mesh .0002
Exterior signs	All assertions in (7.21)
Global derivative mesh	At most .0008 in both coordinates
Outer and negative-part mesh	$D_h \leq .00019$, $S_h \leq .00018$
Outer and negative-part assertions	(7.29), (7.30)
Hessian mesh	d steps .00004, s steps .00008 or smaller
Hessian assertions	(7.32) on every cell

Proposition 7.1 (Meaning of the finite assertions). *Apply the outward interval procedure specified in this section to the radial data (5.7), retaining modes 0 through 38, with the parameters in the table. If every listed assertion holds, its enclosures establish the finite-core premise (6.5), the exterior positivity (7.22), the primitive bound outside the interior of $\mathcal{D}_0$, the negative-part bound for every $d \geq b$, and the convexity bound (7.33), all for the exact infinite-mode functions.*

Proof. The coefficient sum and its lattice tail enclose each retained coefficient. The direct core tests concern only finitely many modes, so they imply the premise needed for (6.10). Equations (7.8)–(7.13) and the midpoint bound prove the node and cell table enclosures by induction through every radial step. The node-separation guard supplies the positive steps and ordered search arrays used by the table and query arguments. The query rules then enclose the exact infinite-mode functions and their first two derivatives, including the constant-region continuation.

The origin expansion and midpoint errors prove the enclosures of λ, μ, K . The signs (7.21) imply the positive tail. The derivative bounds, quadrature errors and bilinear interpolation prove the outer primitive assertion; the Lipschitz estimate proves the negative-part assertion. Finally, (7.31) encloses the complete Hessian, whose positivity and exact stationarity give (7.33). Every stage uses only enclosures established before that stage. $\square$

Finite bounds. The program `certificate.cpp` implements this procedure using `core.hpp` and `grids.hpp`. All three files and an execution wrapper are included in the `verification` directory. One completed guarded run is documented in `verification/README.md`, which binds the code and formula hashes, run metadata, saved integer pairs, and their interpretation below. The wrapper compiles and runs the computation; it does not read the TeX manuscript. The source-to-code and analytic correspondences are separate arguments. The interval denominator is 2^{144} . The table gives outward-rounded projections, while interval assertions use uncoarsened endpoints.

Quantity	Recorded projection
λ	$[-6.7300312, -6.7300275]$
μ	$[\text{.4138830}, \text{.4138838}]$
K	$[-.0034823, -.0034809]$
$B(l)$	$[1.0464643, 1.0464644]$
$f_0(l)$	$[-1.2135966, -1.2135965]$
$f(l)$	$[\text{.0044755}, \text{.0044772}]$
$-\pi B(l) - \lambda/2$	$[\text{.0774490}, \text{.0774509}]$
$m_* - E_0 - E_I$	$\geq .0000120$
Negative-part left side of (7.30)	$\leq -.0004585$
g_{dd} on $\mathcal{D}_0$	≥ 1.0529222
g_{xx} on $\mathcal{D}_0$	$\geq .0128335$
$\det \nabla^2 g$ on $\mathcal{D}_0$	$\geq .0086252$

Here the determinant is enclosed on each individual rectangle before the bounds are combined; it is not computed from separately combined entry ranges. The global derivative bounds, rounded outwards for display, are

$$L_1 \leq 7.4636170, \quad L_2 \leq 2.5091315, \quad L_3 \leq 143.7890461, \quad L_4 \leq 315.1489689.$$

These derivative numbers are projections of the computed hulls, not separate executable threshold assertions. For the identified sources, under the documented runtime assumptions,

the completed run verifies the listed finite assertions using uncoarsened interval endpoints. Proposition 7.1 explains why these assertions imply the required bounds for the exact infinite-mode functions: the interval formulas and range-coverage arguments identify the quantities enclosed, while the analytic truncation estimates supply the omitted modes. Neither the rounded table nor the execution record alone supplies those mathematical correspondences.

Theorem 7.2 (Scalar inequalities from the finite verification). *The exact constants and functions of (5.16) and (5.14) satisfy*

$$K < 0, \quad 2g_d(x) \geq K \quad (d \geq b, x \geq 0), \quad \int_0^\infty \max\{0, -G(d, s)\} ds \leq -K \quad (d \geq b).$$

Proof. The documented finite verification and the correspondences just established supply the hypotheses of Proposition 7.1. The first assertion is part of (7.21). The outer-grid assertion and (7.33) cover all of $\mathcal{D}$. For $d \in [b, .697]$ the positive tail makes the primitive increasing after $x = .65$, so the primitive bound extends to every $x \geq 0$. For $d \geq .697$, $G(d, s) > 0$ for all $s \geq 0$, whence $g_d(x) \geq 0 > K/2$. The negative-part estimate follows from (7.30) on the rectangle; its integrand vanishes outside the rectangle by (7.22). These are exactly (5.18). $\square$

8 The finite-torus and whole-plane bounds

We now apply the geometric comparison to the certified scalar bounds, then pass from square tori to the plane. The square-torus conclusion of Proposition 4.5 already incorporates the separation of minimizers and the passage to arbitrary configurations.

Proof of Theorem 1.2. The radial data of Section 5 satisfy (4.3)–(4.4) with lower distance $d_0 = b = 28209/100000$. Theorem 7.2 gives $K < 0$ and both bounds in (5.18). By Lemma 4.4 at $d_0 = b$ and the primitive identity (5.15), these bounds give (4.14) for every sector with $d \geq b$, including the signed endpoint cases.

Since $b < 1/(2\sqrt{\pi}) = d_*$, as follows from $\pi < 3.1416$, the square-torus conclusion of Proposition 4.5 applies. For every $n \geq 2$ and every configuration of n distinct points it gives, using the exact triangular calibration (5.17),

$$\mathcal{W}_n(h) \geq n(6K + \lambda + 2\pi\mu) = nW(E_\Delta).$$

No separation hypothesis is imposed on this configuration. The same proposition gives the energy penalty $n|c|^2/2$ for a constant harmonic component and preserves the bound under a fixed orthogonal rotation. $\square$

Proof of Theorem 1.1. The periodic triangular field belongs to the admissible class and has finite energy by Lemmas 2.1 and 2.3. Thus $C = W(E_\Delta)$ is a finite real number. Apply Proposition 2.4 to Theorem 1.2 with this C . It gives the lower bound for every field in the stated admissible class and for each fixed cutoff family (1.1). The triangular field attains the lower bound. $\square$

The comparison establishes the minimum of this logarithmic renormalized energy. It does not classify all configurations having that energy per unit area. In particular it supplies no uniqueness or defect-exclusion assertion.

8.1 The spherical logarithmic-energy constant

Proof of Corollary 1.3. Write $\mathcal{A}_m^{\text{BS}}$ and W_{ball} for the admissible class and ball-cutoff energy of Bétermin and Sandier [1, Definitions 2.1–2.4]. Their fields satisfy $\text{div } E = 2\pi(\nu_\Lambda - m)$ and $\text{curl } E = 0$, with the same charge-growth condition and local puncture functional as here. At $m = 1$ the admissible class is our $\mathcal{A}_1$; the local integrability conventions agree by Lemma 2.1. The outer average, however, is over centered balls rather than squares.

The component rotation $j \mapsto F = (j_2, -j_1)$ is a bijection from $\mathcal{C}_1$ to $\mathcal{A}_{1/(2\pi)}^{\text{BS}}$ and preserves the local energy. Thus (2.2) identifies M with $\min_{\mathcal{A}_{1/(2\pi)}^{\text{BS}}} W_{\text{ball}}$. The scaling formula [1, equation (2.4)] gives

$$\min_{\mathcal{A}_1^{\text{BS}}} W_{\text{ball}} = 2\pi \left(M - \frac{1}{4} \log(2\pi) \right). \quad (8.1)$$

Lemma 2.2 bounds every square-cutoff energy $W(E)$ below by the right-hand side. Conversely, periodic averaging applied to the scaled square-periodic sequence in the proof of Proposition 2.4 shows that its square-cutoff energies converge to that value. Theorem 1.1 therefore identifies it with $W(E_\Delta)$. This uses the common ball and square minimum in (2.2), not equality of those two energies for an arbitrary field. Lemma 2.3 does give equality for the periodic field E_Δ , so this covolume-one triangular field attains the minimum in (8.1). Its value also agrees with the periodic lattice notation in Bétermin and Sandier: replacing their gradient ∇H_Λ by the admissible field $-\nabla H_\Lambda$ leaves the local squared-norm functional unchanged.

Theorem 1.5 of [1] gives the form of (1.7) with C in place of C_{BHS} , where

$$C = \frac{1}{\pi} \min_{\mathcal{A}_1^{\text{BS}}} W_{\text{ball}} + \frac{1}{2} \log \pi + \log 2,$$

and states that $C = C_{\text{BHS}}$ exactly when the density-one triangular lattice attains this minimum. Its energy is precisely the ordered-pair sum on the unit sphere in $\mathbb{R}^3$ used here. In particular, the $-n(n-1)\log 2$ term in its passage from the radius-1/2 sphere to the unit sphere contributes both $-n^2 \log 2$ and $+n \log 2$; no further pair-counting or radius factor is needed. The equality clause now gives (1.8) and proves the corollary. $\square$

A Supplementary coefficient and tail estimates

The proof uses the coefficient enclosures and omitted-mode estimates in Sections 6 and 7. This appendix records an independent integer recurrence for the same finite coefficient sums and alternative exact bounds for the same omitted-mode series. These arguments are supplementary: they do not add a hypothesis to the scalar theorem or replace the analytic derivative bounds and finite retained-mode premises used there.

A.1 An independent recurrence for the lattice coefficients

Fix integers $(n, q) \neq (0, 0)$ and write

$$A = 2n + q, \quad N = n^2 + nq + q^2, \quad \zeta = A + i\sqrt{3}q, \quad w = \frac{\zeta}{2}.$$

Thus $w = n + q/2 + i\sqrt{3}q/2$, $N = |w|^2 > 0$, and $|\zeta|^2 = 4N$. A point of the covolume-one triangular lattice is aw , where $a = \sqrt{2/\sqrt{3}}$. For $m \geq 0$, put

$$r_m(n, q) = \text{Re}(\zeta^{6m}).$$

The two numbers $u = \zeta^6$ and $v = \bar{\zeta}^6$ have

$$u + v = 2r_1, \quad uv = (4N)^6,$$

and hence satisfy $X^2 - 2r_1X + (4N)^6 = 0$. Multiplying this equation by u^{m-1} and v^{m-1} , respectively, and adding gives the integer recurrence

$$\begin{aligned} r_0 &= 1, \\ r_1 &= A^6 - 45A^4q^2 + 135A^2q^4 - 27q^6, \\ r_{m+1} &= 2r_1r_m - (4N)^6r_{m-1} \quad (m \geq 1). \end{aligned} \tag{A.1}$$

Here all the quantities on the right are integers. This proves the recurrence for every individual lattice point, without a numerical approximation to a complex power.

The binomial theorem identifies its numerator exactly. Since only even powers of $i\sqrt{3}q$ have nonzero real part,

$$r_m(n, q) = \sum_{j=0}^{3m} (-1)^j \binom{6m}{2j} A^{6m-2j} (3q^2)^j. \tag{A.2}$$

This is the numerator in (7.5), after writing q for its second integer coordinate. Indeed, $w^{-6m} = \bar{w}^{6m}/N^{6m}$, and therefore

$$\text{Re}(w^{-6m}) = \frac{r_m(n, q)}{2^{6m} N^{6m}} = \frac{r_m(n, q)}{(64N^6)^m}. \tag{A.3}$$

Thus the recurrence and the binomial expression agree for every point and every mode, not only for a selected finite list of comparisons.

Grouping by the norm. For a positive integer D , let

$$\mathcal{Q}_D = \{(n, q) \in \mathbb{Z}^2 : |n|, |q| \leq D, (n, q) \neq (0, 0)\}, \quad R_{m,N}(D) = \sum_{\substack{(n,q) \in \mathcal{Q}_D \\ n^2+nq+q^2=N}} r_m(n, q).$$

After forming the numerator for each point, one may combine the equal denominators in (A.3):

$$\sum_{(n,q) \in \mathcal{Q}_D} \text{Re}(w^{-6m}) = \sum_{N \geq 1} \frac{R_{m,N}(D)}{(64N^6)^m}. \tag{A.4}$$

Only finitely many positive norms occur. It is important that the recurrence precede this grouping. For fixed D , summing (A.1) over a norm shell gives

$$R_{m+1,N}(D) = 2 \sum_{\substack{(n,q) \in \mathcal{Q}_D \\ n^2+nq+q^2=N}} r_1(n, q)r_m(n, q) - (4N)^6 R_{m-1,N}(D).$$

In general the first sum cannot be replaced by one value of r_1 times $R_{m,N}(D)$. For example, $(7, 0)$ and $(5, 3)$ both have $N = 49$, but their respective values of r_1 are 7 529 536 and -4 912 064. One may apply the recurrence either pointwise or within groups having the same pair (N, r_1) , and only then combine all terms of a common norm.

For the finite coefficient enclosures, take $D = 256$ when $m = 1$ and $D = 48$ when $2 \leq m \leq 38$. The omitted dimensionless sum has absolute value at most

$$E_{m,D} = 8(4/3)^{3m} \frac{D^{2-6m}}{6m-2}$$

by (6.1). The coefficient c_m is consequently enclosed by adding $[-E_{m,D}, E_{m,D}]$ to the finite sum in (A.4) and then multiplying by the positive factor $a^{-6m}/(6m)$. The square and its omitted tail are unchanged by grouping.

Complex pairs and interval containment. The finite calculation may instead keep both real components. Write

$$\zeta^6 = r_1 + i\sqrt{3}s_1, \quad s_1 = 6A^5q - 60A^3q^3 + 54Aq^5.$$

The multiplication rule

$$(x + i\sqrt{3}y)(x' + i\sqrt{3}y') = (xx' - 3yy') + i\sqrt{3}(xy' + yx')$$

shows that, starting from $(x_0, y_0) = (1, 0)$,

$$(x_{m+1}, y_{m+1}) = (x_m r_1 - 3y_m s_1, x_m s_1 + y_m r_1)$$

represents $\zeta^{6(m+1)}$ exactly. In particular $x_m = r_m$ and its real component obeys (A.1). This is the complex-pair recursion used in the finite calculation, again applied per point before norm grouping. The half square $n > 0$, or $n = 0$ and $q > 0$, contains one point of each pair $(n, q), (-n, -q)$. Because $6m$ is even, the two real numerators agree, which justifies doubling each half-square contribution.

For completeness, grouping also explains an interval containment relation. At fixed denominator $Q = 2^{144}$, let

$$\mathcal{I}_Q(x) = \left[\frac{\lfloor Qx \rfloor}{Q}, \frac{\lceil Qx \rceil}{Q} \right] \quad (x \in \mathbb{Q}).$$

For any finite rational list $x_1, \dots, x_h$, the elementary inequalities

$$\left\lfloor \sum_j Qx_j \right\rfloor \geq \sum_j \lfloor Qx_j \rfloor, \quad \left\lceil \sum_j Qx_j \right\rceil \leq \sum_j \lceil Qx_j \rceil$$

give

$$\mathcal{I}_Q \left(\sum_j x_j \right) \subseteq \sum_j \mathcal{I}_Q(x_j). \quad (\text{A.5})$$

Apply this to the exact rationals in each norm shell. Converting the shell sum once and then summing shells is contained in the enclosure obtained by converting every point separately. Adding the same outward tail interval and multiplying by the same positive scaling enclosure preserve containment, because interval addition, multiplication, and outward rounding are monotone under inclusion. This statement assumes the same primitive rounding rule, tail enclosure, and scaling enclosure on both sides; exact algebra alone does not assert endpoint identity or containment between different choices of those enclosures.

A.2 Alternative exact bounds for the omitted modes

For the allowed indices $k = 234, 240, \dots$, split the weighted summand $(k+1)^4 w_k$ of (6.3) into four positive rationals:

$$\begin{aligned} t_1(k) &= 10^6 30(k+1)^4 k^2 (13/20)^k, \\ t_2(k) &= 10^6 (3 \cdot 10^6)(k+1)^4 k^3 (1/2)^k, \\ t_3(k) &= 10^6 5600(k+1)^4 k^5 (3/5)^k, \\ t_4(k) &= 10^6 (7 \cdot 10^{11})(k+1)^4 k^2 (289/500)^k. \end{aligned}$$

Then

$$\mathfrak{a} = \sum_{j=0}^{\infty} \sum_{i=1}^4 t_i(234 + 6j).$$

This is the same $\mathfrak{a}$ as in (6.6), distinct from the lattice scale a . If

$$(p_i, \beta_i) = (2, 13/20), (3, 1/2), (5, 3/5), (2, 289/500),$$

in the order $i = 1, 2, 3, 4$, the exact consecutive ratios are

$$\rho_i(k) = \frac{t_i(k+6)}{t_i(k)} = \left(\frac{k+7}{k+1}\right)^4 \left(\frac{k+6}{k}\right)^{p_i} \beta_i^6. \quad (\text{A.6})$$

Both rational factors decrease for $k > 0$, so each $\rho_i(k)$ decreases. The four initial ratios, with no rounding, are

i	$\rho_i(234)$
1	$(241/235)^4 (40/39)^2 (13/20)^6$
2	$(241/235)^4 (40/39)^3 (1/2)^6$
3	$(241/235)^4 (40/39)^5 (3/5)^6$
4	$(241/235)^4 (40/39)^2 (289/500)^6$

Here is an exact simultaneous check that all four are less than $1/10$. The integer comparisons

$$9 \cdot 241^4 = 30\,360\,623\,049 < 30\,498\,006\,250 = 10 \cdot 235^4,$$

$$7 \cdot 40^5 = 716\,800\,000 < 721\,793\,592 = 8 \cdot 39^5,$$

$$800 \cdot 13^6 = 3\,861\,447\,200 < 4\,032\,000\,000 = 63 \cdot 20^6$$

give, since $p_i \leq 5$ and $0 < \beta_i \leq 13/20$,

$$\rho_i(234) \leq (241/235)^4 (40/39)^5 (13/20)^6 < \frac{10}{9} \frac{8}{7} (13/20)^6 < \frac{1}{10} \quad (1 \leq i \leq 4). \quad (\text{A.7})$$

Three geometric bounds. Set $T_0 = \sum_{i=1}^4 t_i(234)$. One convenient entirely integer form of its exact comparison is obtained by putting

$$\begin{aligned} C_{234} = & 30 \cdot 234^2 \cdot 325^{234} + 3 \cdot 10^6 \cdot 234^3 \cdot 250^{234} \\ & + 5600 \cdot 234^5 \cdot 300^{234} + 7 \cdot 10^{11} \cdot 234^2 \cdot 289^{234}. \end{aligned}$$

The common denominator 500^{234} gives

$$T_0 = \frac{10^6 235^4 C_{234}}{500^{234}}, \quad 10^{29} 235^4 C_{234} < 156 \cdot 500^{234}.$$

The displayed integer inequality is exactly $T_0 < 156/10^{23}$. In particular,

$$\frac{10}{9} T_0 < \frac{1560}{9 \cdot 10^{23}} < \frac{2}{10^{21}}, \quad (\text{A.8})$$

where the last comparison reduces to $1560 < 1800$.

Let $\rho_{\max} = \max_{1 \leq i \leq 4} \rho_i(234)$, and define three exact rational upper bounds

$$A_{\text{coarse}} = \frac{10}{9} T_0, \quad A_{\text{common}} = \frac{T_0}{1 - \rho_{\max}}, \quad A_{\text{component}} = \sum_{i=1}^4 \frac{t_i(234)}{1 - \rho_i(234)}.$$

Decreasing ratios imply $t_i(234 + 6j) \leq t_i(234) \rho_i(234)^j$. Summing these geometric majorants and using (A.7) proves

$$\mathfrak{a} \leq A_{\text{component}} \leq A_{\text{common}} < A_{\text{coarse}} < \frac{2}{10^{21}}. \quad (\text{A.9})$$

The second inequality holds because $1 - \rho_i(234) \geq 1 - \rho_{\max} > 0$. The common-ratio estimate keeps only the largest initial ratio, whereas the componentwise estimate retains each one; both refine the coarse factor $10/9$.

An exact partial sum and a positive remainder bound. There are exactly 62 allowed indices from 234 through 600, because $600 = 234 + 6 \cdot 61$. Define the finite rational sum

$$P_{600} = \sum_{i=1}^4 \sum_{j=0}^{61} t_i(234 + 6j) \quad (\text{A.10})$$

and the positive rational number

$$R_{606} = \sum_{i=1}^4 \frac{t_i(606)}{1 - \rho_i(606)}, \quad \rho_i(606) = \left(\frac{613}{607}\right)^4 \left(\frac{102}{101}\right)^{p_i} \beta_i^6. \quad (\text{A.11})$$

Every numerator is positive and $0 < \rho_i(606) < \rho_i(234) < 1/10$, so $R_{606} > 0$. Starting the geometric estimate at the next mode 606, rather than at the last included mode 600, gives

$$P_{600} < \mathfrak{a} \leq P_{600} + R_{606}. \quad (\text{A.12})$$

The first inequality is strict because all subsequent summands are positive. The number R_{606} is an upper bound for the remainder, not an assertion that the remainder equals it.

This refinement is no larger than the componentwise bound. For the i -th component put $r = \rho_i(234)$. Then $t_i(606) \leq t_i(234)r^{62}$ and $\rho_i(606) \leq r$, while each term of its partial sum is at most $t_i(234)r^j$. Consequently

$$\sum_{j=0}^{61} t_i(234 + 6j) + \frac{t_i(606)}{1 - \rho_i(606)} \leq t_i(234) \left(\sum_{j=0}^{61} r^j + \frac{r^{62}}{1 - r} \right) = \frac{t_i(234)}{1 - r}.$$

Summing over i places $P_{600} + R_{606}$ below $A_{\text{component}}$ in (A.9). These finite bounds are exact rationals; a rounded display that makes P_{600} and $P_{600} + R_{606}$ look equal does not change (A.12).

Rational propagation to the radial errors. The refined bounds can be substituted directly into the propagation estimates of Section 6.5. They do not replace its analytic mode bounds or the finite premises (6.5): the retained second-order B_1 norm is still bounded by 6000, and the weighted third-order D_i norm by $2 \cdot 10^6$. Recall also that

$$W_* = 4 \sum_{k=234,240,\dots} w_k \leq 4 \sum_{k=234,240,\dots} (k+1)^4 w_k = 4\mathfrak{a}, \quad (\text{A.13})$$

so replacing W_* by $4\mathfrak{a}$ is an upper bound, not an identity. Collecting the integral and mode-norm contributions in (6.8) and using $\pi < 22/7$ gives the rational expression, for $A, W \geq 0$,

$$\begin{aligned} \mathcal{F}(A, W) &= 2\frac{22}{7}(3W + 16A) \\ &\quad + \frac{1}{2}(60000W + 8A + 12(2 \cdot 10^6)W + 48(6000 + W)A \\ &\quad + 8(2 \cdot 10^6)A + 16A^2). \end{aligned} \quad (\text{A.14})$$

The terms are respectively the source integral, the S_2 error, the linear D_i error, the retained and omitted mixed $D_i H_{k_i}$ errors, and the quadratic $D_i D_j$ error. The bound $\pi < 22/7$ and (6.8) give $|f_0 - \tilde{f}_0|_{2,t} \leq \mathcal{F}(\mathfrak{a}, W_*)$. Every coefficient of $\mathcal{F}$ is nonnegative. Thus any one of the rational upper bounds A above, for which $\mathfrak{a} \leq A$, satisfies

$$|f_0 - \tilde{f}_0|_{2,t} \leq \mathcal{F}(\mathfrak{a}, W_*) \leq \mathcal{F}(A, 4A).$$

Direct collection of the rational terms gives the exact identity

$$\mathcal{F}(A, 4A) = A(56\,264\,180 + 104A). \quad (\text{A.15})$$

For example, the source term contributes $176A$; the remaining linear terms contribute $56\,264\,004A$, and the quadratic terms contribute $104A^2$.

Put $A_0 = 2/10^{21}$. Each rational upper bound for $\mathfrak{a}$ above is less than A_0 , and the coefficient in (A.15) increases with A . The exact comparison

$$56\,264\,180 + \frac{208}{10^{21}} < 56\,264\,181 < 6 \cdot 10^7$$

therefore supplies the claimed coefficient bound. More explicitly,

$$\mathcal{F}(A_0, 4A_0) = \frac{112\,528\,360}{10^{21}} + \frac{416}{10^{42}} < \frac{12}{10^{14}}.$$

The order-two norm is $|F|_{2,t} = |F(t)| + |F'(t)| + |F''(t)|/2$. Hence the second derivative, unlike the value and first derivative, requires the factor 2:

$$|f_0'' - \tilde{f}_0''| \leq 2|f_0 - \tilde{f}_0|_{2,t} < \frac{24}{10^{14}} = 2.4 \cdot 10^{-13}.$$

For the angular function, (6.9) similarly yields the entirely rational comparison

$$|e'' - \tilde{e}''| \leq 2(16(2 \cdot 10^6)A_0 + 32A_0^2) = \frac{128\,000\,000}{10^{21}} + \frac{256}{10^{42}} < \frac{13}{10^{14}} = 1.3 \cdot 10^{-13}.$$

The value and first derivative are bounded by their norms without this factor. Thus these rational alternatives preserve the error threshold 10^{-11} for all six quantities in (6.10) on $[b, l]$. For $t \geq l$, the exact continuation (6.11), rather than a uniform absolute error at unbounded radius, remains the applicable statement.

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